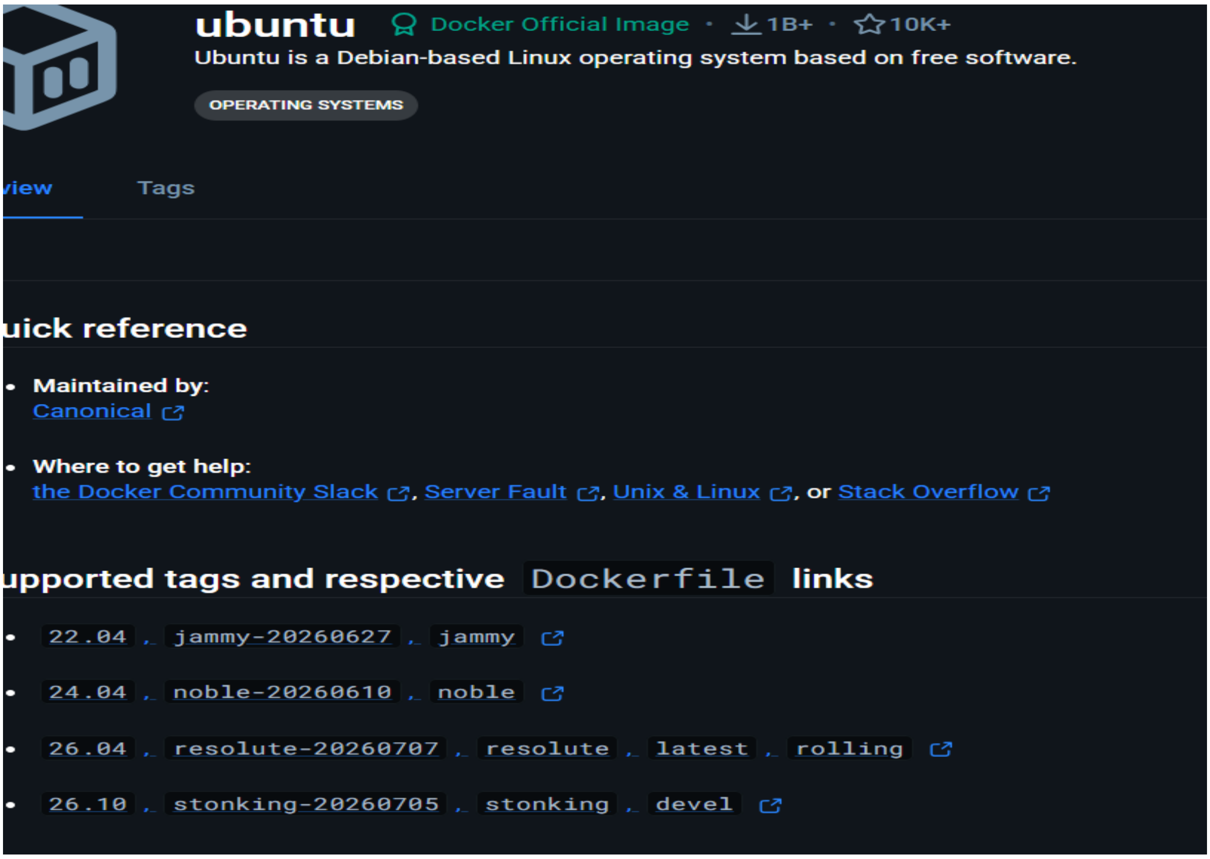
DOCKER ASSIGNMENT

1. **Docker installation**: Install Docker on your local system (Windows, macOS, or Linux). Verify the installation by running the docker --version and docker info commands.

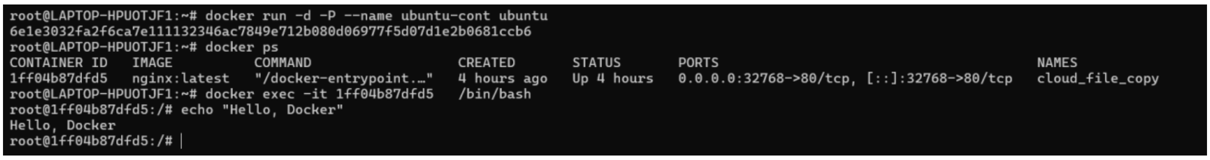
```
oot@LAPTOP-HPUOTJF1:~# docker --version
ocker version 29.1.3, build 29.1.3-0ubuntu4.1
oot@LAPTOP-HPUOTJF1:~# docker info
lient:
Version:    29.1.3
Context:    default
Debug Mode: false
Plugins:
 compose: Docker Compose (Docker Inc.)
   Version:  2.40.3+ds1-0ubuntu1
   Path:     /usr/libexec/docker/cli-plugins/docker-compose
 trust: Manage trust on Docker images (Docker Inc.)
   Version:  29.1.3
   Path:     /usr/libexec/docker/cli-plugins/docker-trust

erver:
Containers: 1
 Running: 1
 Paused: 0
 Stopped: 0
Images: 2
Server Version: 29.1.3
Storage Driver: overlayfs
 driver-type: io.containerd.snapshotter.v1
Logging Driver: json-file
Cgroup Driver: systemd
Cgroup Version: 2
Plugins:
 Volume: local
 Network: bridge host ipvlan macvlan null overlay
 Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
CDI spec directories:
 /etc/cdi
 /var/run/cdi
Swarm: inactive
Runtimes: io.containerd.runc.v2 runc
Default Runtime: runc
Init Binary: docker-init
containerd version:
```

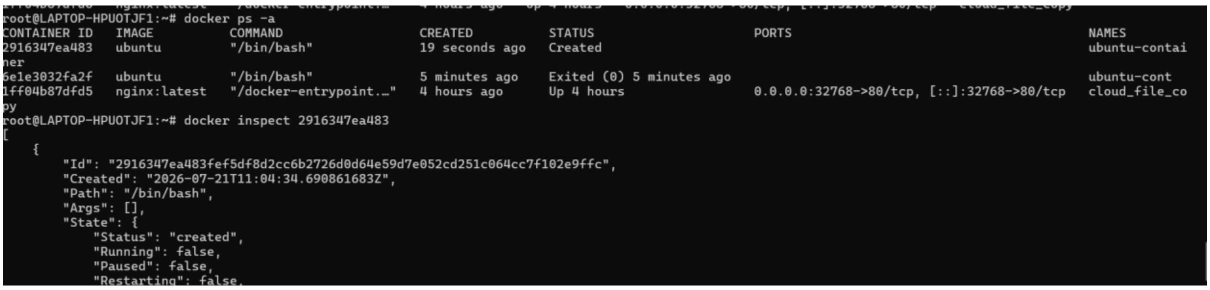
2. **Working with Docker images**: Search for the official Ubuntu image on Docker Hub, pull it to your local system using docker pull, and list all available images on your system using docker images.



3. **Creating a Docker container**: Run a Docker container using the Ubuntu image, and execute a command inside the container (e.g., echo "Hello, Docker!").



4. Inspect the Ubuntu Docker container to ensure it is in the "created" state only.



5. Execute the Ubuntu Docker container and confirm that the container's state is "running."

```
root@LAPTOP-HPUOTJF1:~# docker inspect 1ff04b87dfd5
[
    {
        "Id": "1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88",
        "Created": "2026-07-21T07:15:14.759809887Z",
        "Path": "/docker-entrypoint.sh",
        "Args": [
            "nginx",
            "-g",
            "daemon off;"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
```

6. Launch an HTTP container, verify its state, and then remove the container and image from your local system.

```
root@LAPTOP-HPUOTJF1:~# docker run -d -P --name http-container httpd
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
39d55e966aeb: Pull complete
4f4fb700ef54: Pull complete
69b5b831a361: Pull complete
ebd81352b960: Pull complete
010f7d7e175f: Pull complete
507729664175: Download complete
c72742279dc1: Download complete
Digest: sha256:305fd8326a27a137fedb8900f26375699ab233cc37793f35cf5d7c65671fb252
Status: Downloaded newer image for httpd:latest
e7fc705c2cc58c666e10fe62bdd9f1ff8097413933aabf78ecdaf56cfb6800c9
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS                                     NAMES
e7fc705c2cc5   httpd          "httpd-foreground"       8 seconds ago   Up 7 seconds   0.0.0.0:32769->80/tcp, [::]:32769->80/tcp   http-container
1ff04b87dfd5   nginx:latest   "/docker-entrypoint.…"   4 hours ago     Up 4 hours     0.0.0.0:32768->80/tcp, [::]:32768->80/tcp   cloud_file_copy
root@LAPTOP-HPUOTJF1:~# docker rm -f e7fc
e7fc
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED       STATUS       PORTS                                       NAMES
1ff04b87dfd5   nginx:latest   "/docker-entrypoint.…"   4 hours ago   Up 4 hours   0.0.0.0:32768->80/tcp, [::]:32768->80/tcp   cloud_file_copy
root@LAPTOP-HPUOTJF1:~# docker images
                                                                                        i Info →   U In Use
IMAGE           ID             DISK USAGE   CONTENT SIZE   EXTRA
httpd:latest    305fd8326a27        177MB         47.6MB
nginx:latest    b5a9a3cfc86b        241MB           66MB    U
ubuntu:latest   3131b4cc82a7        161MB         45.3MB    U
root@LAPTOP-HPUOTJF1:~# docker rmi httpd:latest
Untagged: httpd:latest
Deleted: sha256:305fd8326a27a137fedb8900f26375699ab233cc37793f35cf5d7c65671fb252
root@LAPTOP-HPUOTJF1:~# docker images
                                                                                        i Info →   U In Use
IMAGE           ID             DISK USAGE   CONTENT SIZE   EXTRA
nginx:latest    b5a9a3cfc86b        241MB           66MB    U
ubuntu:latest   3131b4cc82a7        161MB         45.3MB    U
root@LAPTOP-HPUOTJF1:~#
```

7. Execute a command to display a filtered list of containers, showing only their short container IDs.

```
root@LAPTOP-HPUOTJF1:~# docker ps -qa
2916347ea483
6e1e3032fa2f
1ff04b87dfd5
root@LAPTOP-HPUOTJF1:~#
```

8. Clone the lancachenet/ubuntu-nginx Git repository locally and build a Docker image from the provided Dockerfile. Be sure to add relevant tags when building the image.

```
root@LAPTOP-HPUOTJF1:~# git clone https://github.com/lancachenet/ubuntu-nginx.git
Cloning into 'ubuntu-nginx'...
remote: Enumerating objects: 149, done.
remote: Counting objects: 100% (57/57), done.
remote: Compressing objects: 100% (36/36), done.
remote: Total 149 (delta 15), reused 41 (delta 11), pack-reused 92 (from 1)
Receiving objects: 100% (149/149), 32.15 KiB | 2.47 MiB/s, done.
Resolving deltas: 100% (30/30), done.
root@LAPTOP-HPUOTJF1:~# ls
1ff                  DevOps_Project  Merge_demo     clone-remote-repo     merge_conflict_demo  revert_demo  student-repo
Amend_repo           GitIgnore-Repo  amend-repo     free-for-dev          project              stash-demo   ubuntu-nginx
Cloud_Ethix_Copyfiles GitWorking     ci-cd-project  githubaction-project  revert-demo          stash1-demo
root@LAPTOP-HPUOTJF1:~# cd ubuntu-nginx/
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# ls
Dockerfile  README.md  goss.yaml  goss_wait.yaml  overlay  run-tests.sh
```

```
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# docker build -t ubuntu-nginx .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  137.7kB
Step 1/7 : FROM lancachenet/ubuntu:latest
latest: Pulling from lancachenet/ubuntu
e5044da18586: Pulling fs layer
199f361bf106: Pulling fs layer
0e5f8475f38b: Pulling fs layer
2b7f00c642c2: Pulling fs layer
e5044da18586: Download complete
2b7f00c642c2: Download complete
199f361bf106: Download complete
0e5f8475f38b: Download complete
```

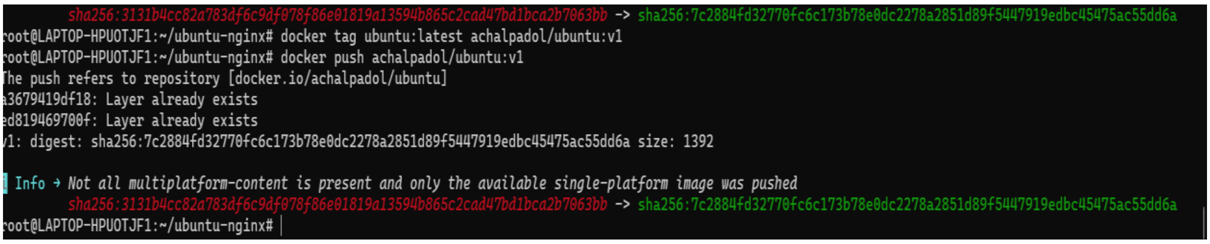
```
 ---> eb1421f302fc
Step 7/7 : EXPOSE 80
 ---> Running in f009dc981031
 ---> Removed intermediate container f009dc981031
 ---> 5102563991c6
Successfully built 5102563991c6
Successfully tagged ubuntu-nginx:latest
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# docker images

IMAGE                       ID             DISK USAGE   CONTENT SIZE   EXTRA
lancachenet/ubuntu:latest   f86a1b63836e        248MB         61.4MB
nginx:latest                b5a9a3cfc86b        241MB           66MB   U
ubuntu-nginx:latest         5102563991c6        315MB         80.4MB
ubuntu:latest               3131b4cc82a7        161MB         45.3MB   U
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx#
```

9. After building the image, run a Docker container using the image and check the webpage using the curl command to ensure it's accessible.

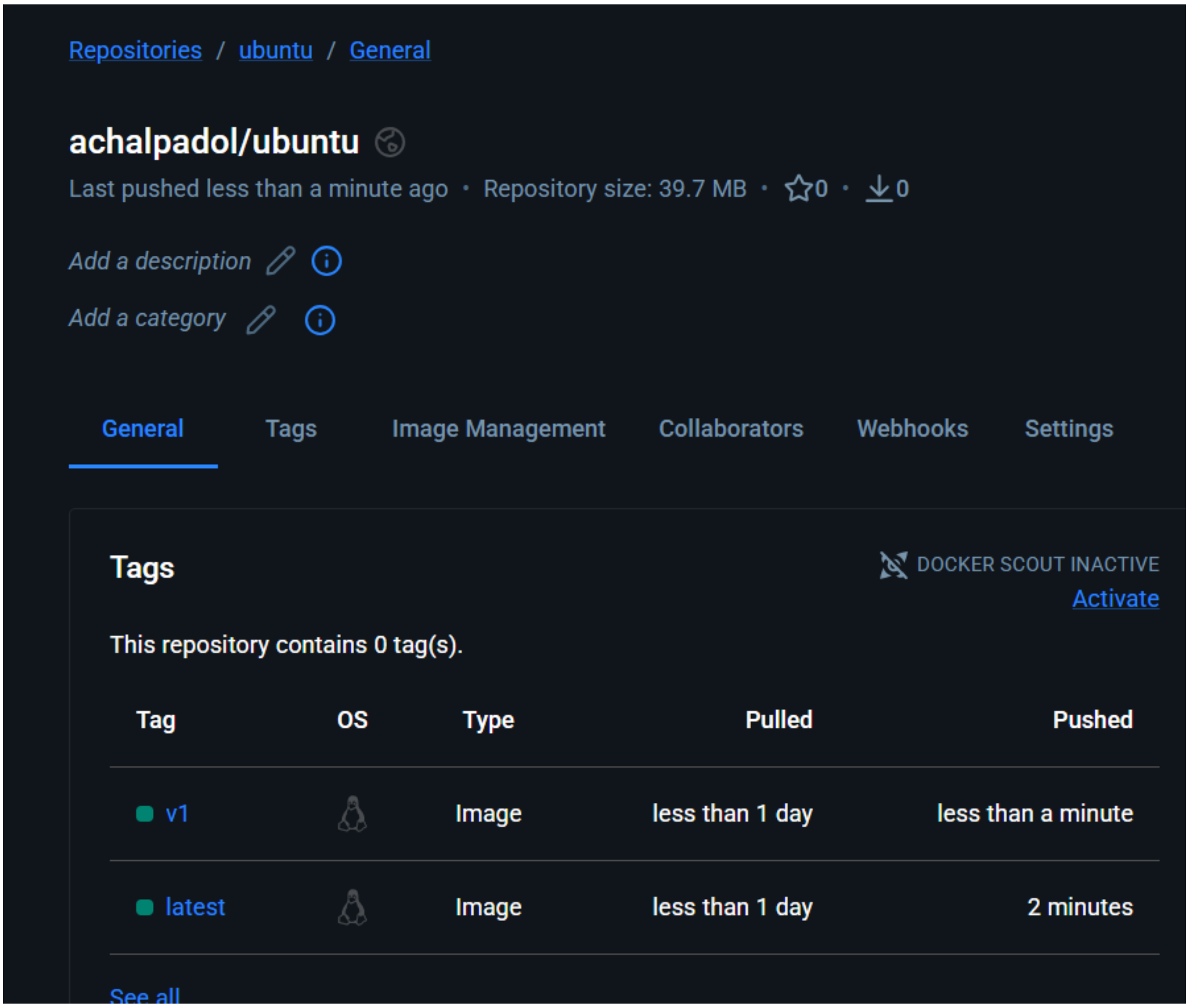
```
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# docker images
                                                                                     Info →   U In Use
IMAGE                       ID             DISK USAGE   CONTENT SIZE   EXTRA
achalpadol/ubuntu:latest    3131b4cc82a7        161MB         45.3MB   U
achalpadol/ubuntu:v1        3131b4cc82a7        161MB         45.3MB   U
lancachenet/ubuntu:latest   f86a1b63836e        248MB         61.4MB
nginx:latest                b5a9a3cfc86b        241MB           66MB   U
ubuntu-nginx:latest         5102563991c6        315MB         80.4MB
ubuntu:latest               3131b4cc82a7        161MB         45.3MB   U
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# docker run -d -P --name webpage-cont ubuntu-nginx:latest
a5f773380c62a2a14dfcde8023ec0bb485674c20e6ac481a6d48df2124fca50e
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED         STATUS         PORTS                                     NAMES
a5f773380c62   ubuntu-nginx:latest   "/bin/bash -e /init/…"   7 seconds ago   Up 6 seconds   0.0.0.0:32770->80/tcp, [::]:32770->80/tcp   webpage-cont
1ff04b87dfd5   nginx:latest          "/docker-entrypoint.…"   4 hours ago     Up 4 hours     0.0.0.0:32768->80/tcp, [::]:32768->80/tcp   cloud_file_copy
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# curl http://localhost:32770
Welcome to the lancachenet/ubuntu-nginx docker container
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx#
```

10. Create a DockerHub account, set up a repository, and tag the locally created Docker image using your_repo_name/application_name:version_number.



```
        sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb -> sha256:7c2884fd32770fc6c173b78e0dc2278a2851d89f5447919edbc45475ac55dd6a
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# docker tag ubuntu:latest achalpadol/ubuntu:v1
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# docker push achalpadol/ubuntu:v1
The push refers to repository [docker.io/achalpadol/ubuntu]
a3679419df18: Layer already exists
ed819469700f: Layer already exists
v1: digest: sha256:7c2884fd32770fc6c173b78e0dc2278a2851d89f5447919edbc45475ac55dd6a size: 1392

i Info → Not all multiplatform-content is present and only the available single-platform image was pushed
        sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb -> sha256:7c2884fd32770fc6c173b78e0dc2278a2851d89f5447919edbc45475ac55dd6a
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx#
```



11. . Launch an Ubuntu container, demonstrate that the escape sequence is functional, and reattach the container after detaching it with the escape sequence. Then, check the container's IP address.

```
oot@LAPTOP-HPUOTJF1:~# docker run -it --name ubuntu-demo ubuntu bash
oot@1b6d1bccb2c0:/# hostname
1b6d1bccb2c0
oot@1b6d1bccb2c0:/# root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS          PORTS                                     NAMES
1b6d1bccb2c0   ubuntu                "bash"                   6 minutes ago    Up 6 minutes                                              ubuntu-demo
a5f773380c62   ubuntu-nginx:latest   "/bin/bash -e /init/…"   13 minutes ago   Up 13 minutes   0.0.0.0:32770->80/tcp, [::]:32770->80/tcp   webpage-cont
1ff04b87dfd5   nginx:latest          "/docker-entrypoint.…"   5 hours ago      Up 5 hours      0.0.0.0:32768->80/tcp, [::]:32768->80/tcp   cloud_file_copy
```

```
},
"NetworkSettings": {
    "SandboxID": "21fb2f76469377e3a818d8a77ce89f9384d4af6a0090e2f98cb8a2ca30e2a8b2",
    "SandboxKey": "/var/run/docker/netns/21fb2f764693",
    "Ports": {},
    "Networks": {
        "bridge": {
            "IPAMConfig": null,
            "Links": null,
            "Aliases": null,
            "DriverOpts": null,
            "GwPriority": 0,
            "NetworkID": "69bcd6cbe2ee95b53de01b7e5351eb2b23995972d7a071fb773351129c780412",
            "EndpointID": "915a2ce2a7896e1cf30d65f84163fa0d39a2f380690f7e990e4a9aea1103731c",
            "Gateway": "172.17.0.1",
            "IPAddress": "172.17.0.4",
            "MacAddress": "4a:34:80:99:9f:7d",
            "IPPrefixLen": 16,
            "IPv6Gateway": "",
            "GlobalIPv6Address": "",
            "GlobalIPv6PrefixLen": 0,
            "DNSNames": null
```

12. Use a Docker command to inspect the hostname and /etc/hosts file of the httpd container.

```
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS          PORTS                                     NAMES
d881f6051b75   httpd                 "httpd-foreground"       4 seconds ago    Up 3 seconds    0.0.0.0:32771->80/tcp, [::]:32771->80/tcp   http-cont
1b6d1bccb2c0   ubuntu                "bash"                   14 minutes ago   Up 14 minutes                                             ubuntu-demo
a5f773380c62   ubuntu-nginx:latest   "/bin/bash -e /init/…"   21 minutes ago   Up 21 minutes   0.0.0.0:32770->80/tcp, [::]:32770->80/tcp   webpage-cont
1ff04b87dfd5   nginx:latest          "/docker-entrypoint.…"   5 hours ago      Up 5 hours      0.0.0.0:32768->80/tcp, [::]:32768->80/tcp   cloud_file_copy
root@LAPTOP-HPUOTJF1:~# docker exec http-cont hostname
d881f6051b75
root@LAPTOP-HPUOTJF1:~# docker exec http-cont cat /etc/hosts
127.0.0.1       localhost
::1     localhost ip6-localhost ip6-loopback
fe00::  ip6-localnet
ff00::  ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
172.17.0.5      d881f6051b75
root@LAPTOP-HPUOTJF1:~#
```

13. Run a Docker command to inspect a container created with the lancachenet/ubuntu-nginx repository, and monitor the container's resource usage and stats with the TOP command.

```
oot@LAPTOP-HPUOTJF1:~# docker inspect webpage-cont

    {
        "Id": "a5f773380c62a2a14dfcde8023ec0bb485674c20e6ac481a6d48df2124fca50e",
        "Created": "2026-07-21T11:33:06.191906718Z",
        "Path": "/bin/bash",
        "Args": [
            "-e",
            "/init/entrypoint",
            "/init/supervisord"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
```

```
CONTAINER ID   NAME           CPU %     MEM USAGE / LIMIT     MEM %     NET I/O         BLOCK I/O         PIDS
a5f773380c62   webpage-cont   0.02%     33.19MiB / 7.581GiB   0.43%     1.9kB / 791B    29.4MB / 28.7kB   15

|
```

```
^Croot@LAPTOP-HPUOTJF1:~# docker top webpage-cont
UID                 PID                 PPID                C                   STIME               TTY                 TIME                CMD
root                8148                8124                0                   11:33               ?                   00:00:00            /usr/bin/python3
 /usr/bin/supervisord -n -c /etc/supervisor/supervisord.conf -e WARN
root                8226                8148                0                   11:33               ?                   00:00:00            /bin/bash /opt/n
ginx/startnginx.sh
root                8227                8226                0                   11:33               ?                   00:00:00            nginx: master pr
ocess /usr/sbin/nginx -g daemon off;
www-data            8228                8227                0                   11:33               ?                   00:00:00            nginx: worker pr
ocess
www-data            8229                8227                0                   11:33               ?                   00:00:00            nginx: worker pr
ocess
www-data            8230                8227                0                   11:33               ?                   00:00:00            nginx: worker pr
ocess
www-data            8231                8227                0                   11:33               ?                   00:00:00            nginx: worker pr
ocess
www-data            8232                8227                0                   11:33               ?                   00:00:00            nginx: worker pr
ocess
www-data            8233                8227                0                   11:33               ?                   00:00:00            nginx: worker pr
ocess
www-data            8234                8227                0                   11:33               ?                   00:00:00            nginx: worker pr
ocess
www-data            8235                8227                0                   11:33               ?                   00:00:00            nginx: worker pr
ocess
www-data            8236                8227                0                   11:33               ?                   00:00:00            nginx: worker pr
ocess
www-data            8237                8227                0                   11:33               ?                   00:00:00            nginx: worker pr
ocess
www-data            8238                8227                0                   11:33               ?                   00:00:00            nginx: worker pr
ocess
www-data            8239                8227                0                   11:33               ?                   00:00:00            nginx: worker pr
ocess
root@LAPTOP-HPUOTJF1:~# |
```

14. Start a new Nginx container and review the logs for both Nginx containers.

```
root@LAPTOP-HPUOTJF1:~# docker run -d -P --name nginx-cont1 nginx:latest
41befcc7c7078a9c5fc083b68ad7f91a5179b73c1913d1560ca12d290e1817d6
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS          PORTS                                     NAMES
41befcc7c707   nginx:latest          "/docker-entrypoint.…"   4 seconds ago    Up 4 seconds    0.0.0.0:32772->80/tcp, [::]:32772->80/tcp   nginx-cont1
d881f6051b75   httpd                 "httpd-foreground"       10 minutes ago   Up 10 minutes   0.0.0.0:32771->80/tcp, [::]:32771->80/tcp   http-cont
1b6d1bccb2c0   ubuntu                "bash"                   24 minutes ago   Up 24 minutes                                             ubuntu-demo
a5f773380c62   ubuntu-nginx:latest   "/bin/bash -e /init/…"   31 minutes ago   Up 31 minutes   0.0.0.0:32770->80/tcp, [::]:32770->80/tcp   webpage-cont
1ff04b87dfd5   nginx:latest          "/docker-entrypoint.…"   5 hours ago      Up 5 hours      0.0.0.0:32768->80/tcp, [::]:32768->80/tcp   cloud_file_copy
root@LAPTOP-HPUOTJF1:~# docker logs cloud_file_copy
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/21 07:15:15 [notice] 1#1: using the "epoll" event method
2026/07/21 07:15:15 [notice] 1#1: nginx/1.31.2
2026/07/21 07:15:15 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/21 07:15:15 [notice] 1#1: OS: Linux 6.6.114.1-microsoft-standard-WSL2
2026/07/21 07:15:15 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
2026/07/21 07:15:15 [notice] 1#1: start worker processes
2026/07/21 07:15:15 [notice] 1#1: start worker process 29
2026/07/21 07:15:15 [notice] 1#1: start worker process 30
2026/07/21 07:15:15 [notice] 1#1: start worker process 31
2026/07/21 07:15:15 [notice] 1#1: start worker process 32
2026/07/21 07:15:15 [notice] 1#1: start worker process 33
2026/07/21 07:15:15 [notice] 1#1: start worker process 34
2026/07/21 07:15:15 [notice] 1#1: start worker process 35
2026/07/21 07:15:15 [notice] 1#1: start worker process 36
2026/07/21 07:15:15 [notice] 1#1: start worker process 37
2026/07/21 07:15:15 [notice] 1#1: start worker process 38
2026/07/21 07:15:15 [notice] 1#1: start worker process 39
```

```
026/07/21 07:15:15 [notice] 1#1: start worker process 40
oot@LAPTOP-HPUOTJF1:~# docker logs nginx-cont1
docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
0-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
0-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
docker-entrypoint.sh: Configuration complete; ready for start up
026/07/21 12:04:41 [notice] 1#1: using the "epoll" event method
026/07/21 12:04:41 [notice] 1#1: nginx/1.31.2
026/07/21 12:04:41 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
026/07/21 12:04:41 [notice] 1#1: OS: Linux 6.6.114.1-microsoft-standard-WSL2
026/07/21 12:04:41 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
026/07/21 12:04:41 [notice] 1#1: start worker processes
026/07/21 12:04:41 [notice] 1#1: start worker process 29
026/07/21 12:04:41 [notice] 1#1: start worker process 30
026/07/21 12:04:41 [notice] 1#1: start worker process 31
026/07/21 12:04:41 [notice] 1#1: start worker process 32
026/07/21 12:04:41 [notice] 1#1: start worker process 33
026/07/21 12:04:41 [notice] 1#1: start worker process 34
026/07/21 12:04:41 [notice] 1#1: start worker process 35
026/07/21 12:04:41 [notice] 1#1: start worker process 36
026/07/21 12:04:41 [notice] 1#1: start worker process 37
026/07/21 12:04:41 [notice] 1#1: start worker process 38
026/07/21 12:04:41 [notice] 1#1: start worker process 39
026/07/21 12:04:41 [notice] 1#1: start worker process 40
oot@LAPTOP-HPUOTJF1:~# |
```

15. Examine system events, filtering them by date and the last 30 minutes. Additionally, apply two filters simultaneously using name and event.

```
root@LAPTOP-HPUOTJF1:~# docker events --since 30m
2026-07-21T11:54:10.406240831Z image pull httpd:latest (name=httpd)
2026-07-21T11:54:10.531494248Z container create d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadc7d44e61ae (image=httpd, name=http-cont)
2026-07-21T11:54:10.697209520Z network connect 69bcd6cbe2ee95b53de01b7e5351eb2b23995972d7a071fb773351129c780412 (container=d881f6051b755b31b4bf14503f04ad1ee
0ecbaddc77c5e0a912dadc7d44e61ae, name=bridge, type=bridge)
2026-07-21T11:54:10.707743433Z container start d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadc7d44e61ae (image=httpd, name=http-cont)
2026-07-21T11:55:46.403396577Z container exec_create: hostname  d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadc7d44e61ae (execID=4777c4c17fbae0c4181
4e0d50dff71fc86b37ba4dd9e9a6710542f5ae5d2094d, image=httpd, name=http-cont)
2026-07-21T11:55:46.403960046Z container exec_start: hostname  d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadc7d44e61ae (execID=4777c4c17fbae0c41814
e0d50dff71fc86b37ba4dd9e9a6710542f5ae5d2094d, image=httpd, name=http-cont)
2026-07-21T11:55:46.443713592Z container exec_die d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadc7d44e61ae (execID=4777c4c17fbae0c41814e0d50dff71fc8
6b37ba4dd9e9a6710542f5ae5d2094d, exitCode=0, image=httpd, name=http-cont)
2026-07-21T11:56:01.441429119Z container exec_create: cat /etc/hosts d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadc7d44e61ae (execID=2497a51589568d
c2fc36f0073ea57b3905e016ef63824a62e42880f173d4f6dc, image=httpd, name=http-cont)
2026-07-21T11:56:01.441798394Z container exec_start: cat /etc/hosts d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadc7d44e61ae (execID=2497a51589568dc
2fc36f0073ea57b3905e016ef63824a62e42880f173d4f6dc, image=httpd, name=http-cont)
2026-07-21T11:56:01.473823015Z container exec_die d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadc7d44e61ae (execID=2497a51589568dc2fc36f0073ea57b390
5e016ef63824a62e42880f173d4f6dc, exitCode=0, image=httpd, name=http-cont)
2026-07-21T12:01:20.323226933Z container top a5f773380c62a2a14dfcde8023ec0bb485674c20e6ac481a6d48df2124fca50e (image=ubuntu-nginx:latest, maintainer=LanCach
e.Net Team <team@lancache.net>, name=webpage-cont, org.opencontainers.image.version=24.04)
2026-07-21T12:04:41.386710709Z container create 41befcc7c7078a9c5fc083b68ad7f91a5179b73c1913d1560ca12d290e1817d6 (image=nginx:latest, maintainer=NGINX Docke
r Maintainers <docker-maint@nginx.com>, name=nginx-cont1)
2026-07-21T12:04:41.542243227Z network connect 69bcd6cbe2ee95b53de01b7e5351eb2b23995972d7a071fb773351129c780412 (container=41befcc7c7078a9c5fc083b68ad7f91a5
179b73c1913d1560ca12d290e1817d6, name=bridge, type=bridge)
2026-07-21T12:04:41.553329927Z container start 41befcc7c7078a9c5fc083b68ad7f91a5179b73c1913d1560ca12d290e1817d6 (image=nginx:latest, maintainer=NGINX Docker
 Maintainers <docker-maint@nginx.com>, name=nginx-cont1)
```

```
root@LAPTOP-HPUOTJF1:~# docker events --since "2026-07-21T00:00:00"
2026-07-21T07:15:14.798236122Z container create 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (image=nginx:latest, maintainer=NGINX Docke
r Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:15:14.978495810Z network connect 69bcd6cbe2ee95b53de01b7e5351eb2b23995972d7a071fb773351129c780412 (container=1ff04b87dfd56c6bf2e193810432698c0
b355b2395ed482addcbcac32d67eb88, name=bridge, type=bridge)
2026-07-21T07:15:14.991353865Z container start 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (image=nginx:latest, maintainer=NGINX Docker
 Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:16:48.442258188Z container exec_create: /bin/bash  1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (execID=cde3029b4c9a43ae5b
a5a758fe23dc86309b575114288b04399757c3fc33c9e1, image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:16:48.443807868Z container exec_start: /bin/bash  1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (execID=cde3029b4c9a43ae5ba
5a758fe23dc86309b575114288b04399757c3fc33c9e1, image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:16:56.122985154Z container exec_die 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (execID=cde3029b4c9a43ae5ba5a758fe23dc863
09b575114288b04399757c3fc33c9e1, exitCode=0, image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:18:19.109803014Z container exec_create: /bin/bash  1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (execID=2035942d01dd3ac298
9ece75937b71a6320c1105ff374bb445d174f26cd7b540, image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:18:19.110769765Z container exec_start: /bin/bash  1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (execID=2035942d01dd3ac2989
ece75937b71a6320c1105ff374bb445d174f26cd7b540, image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:19:34.384579298Z container exec_die 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (execID=2035942d01dd3ac2989ece75937b71a63
20c1105ff374bb445d174f26cd7b540, exitCode=0, image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:23:02.631601892Z container extract-to-dir 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (image=nginx:latest, maintainer=NGI
NX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:23:17.301740933Z container exec_create: /bin/bash  1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (execID=9e0c7e33508f37c5b5
```

```
root@LAPTOP-HPUOTJF1:~# docker events --since 30m \
  --filter container=http-cont \
  --filter event=start
2026-07-21T11:54:10.707743433Z container start d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadc7d44e61ae (image=httpd, name=http-cont)
```

16. Create a Docker container using the Ubuntu image, assign it a meaningful name, and set the hostname to "cloudethix.com." Stop or kill the container, check its exit code, and examine the container's log for more information.

```
ubuntu-cloud
root@LAPTOP-HPUOTJF1:~# docker exec -it ubuntu-cloud /bin/bash
root@cloudethix:/# hostname
cloudethix.com
```

```
root@LAPTOP-HPUOTJF1:~# docker run -dit \
  --name ubuntu-cloud \
  --hostname cloudethix.com \
  ubuntu bash
dcdbfed91d202f3d6cbf857e0ab01c24613b46cd78d3c59b450d996c33b21815
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS          PORTS                                       NAMES
dcdbfed91d20   ubuntu                "bash"                   4 seconds ago    Up 4 seconds                                                ubuntu-cloud
41befcc7c707   nginx:latest          "/docker-entrypoint.…"   16 minutes ago   Up 16 minutes   0.0.0.0:32772->80/tcp, [::]:32772->80/tcp   nginx-cont1
d881f6051b75   httpd                 "httpd-foreground"       27 minutes ago   Up 27 minutes   0.0.0.0:32771->80/tcp, [::]:32771->80/tcp   http-cont
1b6d1bccb2c0   ubuntu                "bash"                   41 minutes ago   Up 41 minutes                                               ubuntu-demo
a5f773380c62   ubuntu-nginx:latest   "/bin/bash -e /init/…"   48 minutes ago   Up 48 minutes   0.0.0.0:32770->80/tcp, [::]:32770->80/tcp   webpage-cont
1ff04b87dfd5   nginx:latest          "/docker-entrypoint.…"   5 hours ago      Up 5 hours      0.0.0.0:32768->80/tcp, [::]:32768->80/tcp   cloud_file_copy
root@LAPTOP-HPUOTJF1:~# docker stop dcd
dcd
root@LAPTOP-HPUOTJF1:~# docker ps -a
CONTAINER ID   IMAGE                 COMMAND                  CREATED              STATUS                      PORTS
  NAMES
dcdbfed91d20   ubuntu                "bash"                   About a minute ago   Exited (137) 5 seconds ago
  ubuntu-cloud
```

```
Hello from CloudEthix
root@LAPTOP-HPUOTJF1:~# docker logs ubuntu-log
Hello from CloudEthix
root@LAPTOP-HPUOTJF1:~#
```

17. Instantiate and start an Ubuntu container with a random name, then rename it to "Ubuntu01." Launch another container named "Ubuntu02" and check the hostname of both containers. Pause and un-pause the "Ubuntu02" container, and stop, start, and restart the "Ubuntu01" container. Inspect the stats and system events of both containers, then kill and delete them, ensuring that directories created by the containers are removed.

```
root@LAPTOP-HPUOTJF1:~# docker run -dit ubuntu bash
ac6a1c7a8a135b51ae2c79d5db99e247e1dda24d8ae0930937ab5c2c7ce03209
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED         STATUS         PORTS     NAMES
ac6a1c7a8a13   ubuntu    "bash"    4 seconds ago   Up 3 seconds             pedantic_he
rtz
```

```
root@LAPTOP-HPUOTJF1:~# docker rename  pedantic_hertz Ubuntu01
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED              STATUS              PORTS     NAMES
ac6a1c7a8a13   ubuntu    "bash"    About a minute ago   Up About a minute             Ubuntu01
```

```
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED         STATUS         PORTS     NAMES
b12d575cf077   ubuntu    "bash"    4 seconds ago   Up 4 seconds             Ubuntu02
ac6a1c7a8a13   ubuntu    "bash"    3 minutes ago   Up 3 minutes             Ubuntu01
```

```
copy
root@LAPTOP-HPUOTJF1:~# docker exec Ubuntu01 hostname
ac6a1c7a8a13
root@LAPTOP-HPUOTJF1:~# docker exec Ubuntu02 hostname
b12d575cf077
root@LAPTOP-HPUOTJF1:~# |
```

```
Ubuntu02
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED         STATUS                  PORTS     NAMES
b12d575cf077   ubuntu    "bash"    2 minutes ago   Up 2 minutes (Paused)             Ubuntu
02
```

```
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED         STATUS         PORTS     NAMES
b12d575cf077   ubuntu    "bash"    2 minutes ago   Up 2 minutes             Ubuntu02
ac6a1c7a8a13   ubuntu    "bash"    6 minutes ago   Up 6 minutes             Ubuntu01
```

```
root@LAPTOP-HPUOTJF1:~# docker stop ac6
^Croot@LAPTOP-HPUOTJF1:~# ^C
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED             STATUS              PORTS                                     NAMES
b12d575cf077   ubuntu                "bash"                   4 minutes ago       Up 4 minutes                                                  Ubuntu02
dcdbfed91d20   ubuntu                "bash"                   21 minutes ago      Up 16 minutes                                                 ubuntu-clou
d
41befcc7c707   nginx:latest          "/docker-entrypoint.…"   38 minutes ago      Up 38 minutes       0.0.0.0:32772->80/tcp, [::]:32772->80/tcp   nginx-cont1
d881f6051b75   httpd                 "httpd-foreground"       49 minutes ago      Up 49 minutes       0.0.0.0:32771->80/tcp, [::]:32771->80/tcp   http-cont
1b6d1bccb2c0   ubuntu                "bash"                   About an hour ago   Up About an hour                                              ubuntu-demo
a5f773380c62   ubuntu-nginx:latest   "/bin/bash -e /init/…"   About an hour ago   Up About an hour    0.0.0.0:32770->80/tcp, [::]:32770->80/tcp   webpage-con
t
1ff04b87dfd5   nginx:latest          "/docker-entrypoint.…"   5 hours ago         Up 5 hours          0.0.0.0:32768->80/tcp, [::]:32768->80/tcp   cloud_file_
copy
root@LAPTOP-HPUOTJF1:~# docker ps -a
CONTAINER ID   IMAGE                 COMMAND                  CREATED             STATUS                          PORTS
NAMES
b12d575cf077   ubuntu                "bash"                   4 minutes ago       Up 4 minutes
Ubuntu02
ac6a1c7a8a13   ubuntu                "bash"                   8 minutes ago       Exited (137) 18 seconds ago
Ubuntu01
```

```
 cloud_file_copy
root@LAPTOP-HPUOTJF1:~# docker start ac6
ac6
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED          STATUS          PORTS     NAMES
b12d575cf077   ubuntu    "bash"    6 minutes ago    Up 6 minutes              Ubuntu02
ac6a1c7a8a13   ubuntu    "bash"    9 minutes ago    Up 7 seconds              Ubuntu01
dcdbfed91d20   ubuntu    "bash"    23 minutes ago   Up 17 minutes             ubuntu-clou
d
```

```
root@LAPTOP-HPUOTJF1:~# docker restart Ubuntu01
Ubuntu01
root@LAPTOP-HPUOTJF1:~# |
```

Stats of the both containers

```
CONTAINER ID   NAME       CPU %     MEM USAGE / LIMIT     MEM %     NET I/O          BLOCK I/O   PIDS
ac6a1c7a8a13   Ubuntu01   0.00%     884KiB / 7.581GiB     0.01%     726B / 126B      0B / 0B     1
b12d575cf077   Ubuntu02   0.00%     952KiB / 7.581GiB     0.01%     1.19kB / 126B    0B / 0B     1
|
```

Events of the ubuntu01

Run this cmd on the another terminal then check the events on the first terminal.

```
[sudo: authenticate] Password:
root@LAPTOP-HPUOTJF1:~# docker stop Ubuntu01
^Croot@LAPTOP-HPUOTJF1:~# docker start Ubuntu01
Ubuntu01
root@LAPTOP-HPUOTJF1:~# docker restart Ubuntu01
Ubuntu01
root@LAPTOP-HPUOTJF1:~# docker pause Ubuntu02
Ubuntu02
root@LAPTOP-HPUOTJF1:~# docker unpause Ubuntu02
Ubuntu02
root@LAPTOP-HPUOTJF1:~# docker restart Ubuntu02
Ubuntu02
root@LAPTOP-HPUOTJF1:~#
```

```
root@LAPTOP-HPUOTJF1:~# docker events --filter container=Ubuntu01
2026-07-21T12:51:32.821834695Z container kill ac6a1c7a8a135b51ae2c79d5db99e247e1dda24d8ae0930937ab5c2c7ce03209 (image=ubuntu, name=Ubuntu01, org.open
ers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

Ubuntu is a Debian-based Linux operating system that runs from the desktop to the cloud, to all your internet connected things.
It is the world's most popular operating system across public clouds and OpenStack clouds.
It is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu can run your containers at scale.
Fast, secure and simple, Ubuntu powers millions of PCs worldwide.
, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04, signal=15)
2026-07-21T12:51:42.837129595Z container kill ac6a1c7a8a135b51ae2c79d5db99e247e1dda24d8ae0930937ab5c2c7ce03209 (image=ubuntu, name=Ubuntu01, org.open
ers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

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Fast, secure and simple, Ubuntu powers millions of PCs worldwide.
, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04, signal=9)
2026-07-21T12:51:43.009102324Z container stop ac6a1c7a8a135b51ae2c79d5db99e247e1dda24d8ae0930937ab5c2c7ce03209 (image=ubuntu, name=Ubuntu01, org.open
ers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

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Fast, secure and simple, Ubuntu powers millions of PCs worldwide.
, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04)
2026-07-21T12:51:43.011616289Z container die ac6a1c7a8a135b51ae2c79d5db99e247e1dda24d8ae0930937ab5c2c7ce03209 (execDuration=337, exitCode=137, image=
 name=Ubuntu01, org.opencontainers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image ma
d by Canonical
```

Event of the ubuntu02

```
^Croot@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# docker events --filter container=Ubuntu02

2026-07-21T12:58:19.113279016Z container pause b12d575cf07732453db744952f2b728e358451e9074bc259adbc74206ba616b3 (image=ubuntu, name=Ubuntu02, org.opencontai
ners.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

Ubuntu is a Debian-based Linux operating system that runs from the desktop to the cloud, to all your internet connected things.
It is the world's most popular operating system across public clouds and OpenStack clouds.
It is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu can run your containers at scale.
Fast, secure and simple, Ubuntu powers millions of PCs worldwide.
, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04)
2026-07-21T12:58:29.845691035Z container unpause b12d575cf07732453db744952f2b728e358451e9074bc259adbc74206ba616b3 (image=ubuntu, name=Ubuntu02, org.opencont
ainers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

Ubuntu is a Debian-based Linux operating system that runs from the desktop to the cloud, to all your internet connected things.
It is the world's most popular operating system across public clouds and OpenStack clouds.
It is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu can run your containers at scale.
Fast, secure and simple, Ubuntu powers millions of PCs worldwide.
, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04)
2026-07-21T12:58:36.867574193Z container kill b12d575cf07732453db744952f2b728e358451e9074bc259adbc74206ba616b3 (image=ubuntu, name=Ubuntu02, org.opencontain
ers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

Ubuntu is a Debian-based Linux operating system that runs from the desktop to the cloud, to all your internet connected things.
```

18. Run four HTTPD Docker containers with distinct, meaningful names, and apply restart policies (NO, On-Failure, Always, and Unless-Stopped) to each of the four containers, respectively. Demonstrate that the restart policies function as expected.
    - Four containers are created with different name

```
4f2ed216677d01fb8e0ad38e62a990d5f91542b72b652c487181ef3a0233a0bc
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND              CREATED              STATUS              PORTS                                       NAMES
4f2ed216677d   httpd     "httpd-foreground"   5 seconds ago        Up 4 seconds        0.0.0.0:32771->80/tcp, [::]:32771->80/tcp   httpd-unless
977d5e5c948f   httpd     "httpd-foreground"   23 seconds ago       Up 22 seconds       0.0.0.0:32770->80/tcp, [::]:32770->80/tcp   httpd-always
db96e974a76b   httpd     "httpd-foreground"   54 seconds ago       Up 53 seconds       0.0.0.0:32769->80/tcp, [::]:32769->80/tcp   httpd-onfailure
f67caab69cc5   httpd     "httpd-foreground"   About a minute ago   Up About a minute   0.0.0.0:32768->80/tcp, [::]:32768->80/tcp   httpd-no
```

1. --restart=no

No restart policy will not start container automatically

```
root@LAPTOP-HPUOTJF1:~# docker stop httpd-no
httpd-no
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND              CREATED              STATUS              PORTS                                     NAMES
4f2ed216677d   httpd     "httpd-foreground"   About a minute ago   Up About a minute   0.0.0.0:32771->80/tcp, [::]:32771->80/tcp   httpd-unless
977d5e5c948f   httpd     "httpd-foreground"   About a minute ago   Up About a minute   0.0.0.0:32770->80/tcp, [::]:32770->80/tcp   httpd-always
db96e974a76b   httpd     "httpd-foreground"   About a minute ago   Up About a minute   0.0.0.0:32769->80/tcp, [::]:32769->80/tcp   httpd-onfailure
```

2. 2. --restart=on-failure

   This policy only restarts the container if the main process exits with a **non-zero exit code** (an error).

```
root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# docker stop httpd-onfailure
httpd-onfailure
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND              CREATED         STATUS         PORTS                                       NAMES
4f2ed216677d   httpd     "httpd-foreground"   4 minutes ago   Up 4 minutes   0.0.0.0:32771->80/tcp, [::]:32771->80/tcp   httpd-unless
977d5e5c948f   httpd     "httpd-foreground"   5 minutes ago   Up 5 minutes   0.0.0.0:32770->80/tcp, [::]:32770->80/tcp   httpd-always
root@LAPTOP-HPUOTJF1:~# docker ps -a
CONTAINER ID   IMAGE               COMMAND              CREATED         STATUS                      PORTS                                       NAMES
4f2ed216677d   httpd               "httpd-foreground"   5 minutes ago   Up 5 minutes                0.0.0.0:32771->80/tcp, [::]:32771->80/tcp   httpd-unless
977d5e5c948f   httpd               "httpd-foreground"   5 minutes ago   Up 5 minutes                0.0.0.0:32770->80/tcp, [::]:32770->80/tcp   httpd-always
db96e974a76b   httpd               "httpd-foreground"   5 minutes ago   Exited (0) 17 seconds ago                                               httpd-onfailure
```

   will not restart it, because docker stop sends a graceful stop signal, resulting in exit code 0.

3. --restart=always

```
root@LAPTOP-HPUOTJF1:~# docker stop httpd-always
httpd-always
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND              CREATED          STATUS          PORTS                                       NAMES
4f2ed216677d   httpd     "httpd-foreground"   10 minutes ago   Up 10 minutes   0.0.0.0:32771->80/tcp, [::]:32771->80/tcp   httpd-unless
root@LAPTOP-HPUOTJF1:~# docker inspect -f '{{.HostConfig.RestartPolicy.Name}}'
docker: 'docker inspect' requires at least 1 argument

Usage:  docker inspect [OPTIONS] NAME|ID [NAME|ID...]

See 'docker inspect --help' for more information
root@LAPTOP-HPUOTJF1:~# docker inspect -f '{{.HostConfig.RestartPolicy.Name}}' httpd-always
always
root@LAPTOP-HPUOTJF1:~# sudo systemctl restart docker
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND              CREATED          STATUS         PORTS                                       NAMES
4f2ed216677d   httpd     "httpd-foreground"   13 minutes ago   Up 3 seconds   0.0.0.0:32768->80/tcp, [::]:32768->80/tcp   httpd-unless
977d5e5c948f   httpd     "httpd-foreground"   13 minutes ago   Up 3 seconds   0.0.0.0:32769->80/tcp, [::]:32769->80/tcp   httpd-always
root@LAPTOP-HPUOTJF1:~#
```

4. --restart=unless-stopped

   Restart the container automatically unless it was manually stopped by the user

```
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND              CREATED          STATUS         PORTS                                       NAMES
977d5e5c948f   httpd     "httpd-foreground"   17 minutes ago   Up 4 seconds   0.0.0.0:32768->80/tcp, [::]:32768->80/tcp   httpd-always
root@LAPTOP-HPUOTJF1:~# docker inspect -f '{{.HostConfig.RestartPolicy.Name}}' httpd-unless
unless-stopped
root@LAPTOP-HPUOTJF1:~#
```

19. Launch an NGINX container with a meaningful name and expose it on the host's port 80. Create an "index.html" file containing the text "Hello there, Let's be the Team CloudEthiX," and copy the file to the container's "/usr/share/nginx/html/" location. Access the container in a browser to verify that the webpage displays correctly.

```
unless-stopped
root@LAPTOP-HPUOTJF1:~# docker run -d --name nginx-web -p 80:80 nginx
1b33c35e955b7a877bea3ec1907332b117feaa7021c5717d105f1ec76735c2a8
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS         PORTS                                       NAMES
1b33c35e955b   nginx     "/docker-entrypoint.…"   4 seconds ago    Up 3 seconds   0.0.0.0:80->80/tcp, [::]:80->80/tcp         nginx-web
977d5e5c948f   httpd     "httpd-foreground"       21 minutes ago   Up 4 minutes   0.0.0.0:32768->80/tcp, [::]:32768->80/tcp   httpd-always
root@LAPTOP-HPUOTJF1:~# echo "Hello there, Let's be the Team CloudEthiX" > index.html
root@LAPTOP-HPUOTJF1:~# cat index.html
Hello there, Let's be the Team CloudEthiX
root@LAPTOP-HPUOTJF1:~# docker cp index.html nginx-web:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to nginx-web:/usr/share/nginx/html/index.html
root@LAPTOP-HPUOTJF1:~# docker exec nginx-web cat /usr/share/nginx/html/index.html
Hello there, Let's be the Team CloudEthiX
root@LAPTOP-HPUOTJF1:~#
```

20. Pull the Redis image and tag it as "yourname_redis:version 1." Create two additional tags, "yourname_redis:version 2" and "yourname_redis:version 3." Run two Docker containers from the tagged images named "redis01" and "redis02," then delete the "yourname_redis:version 2" image.

```
IMAGE                       ID             DISK USAGE   CONTENT SIZE   EXTRA
achalpadol/ubuntu:latest    3131b4cc82a7        161MB         45.3MB    U
achalpadol/ubuntu:v1        3131b4cc82a7        161MB         45.3MB    U
httpd:latest                305fd8326a27        177MB         47.6MB    U
lancachenet/ubuntu:latest   f86a1b63836e        248MB         61.4MB
nginx:latest                b5a9a3cfc86b        241MB           66MB    U
redis:latest                234c902a2db4        208MB         56.3MB
ubuntu-nginx:latest         5102563991c6        315MB         80.4MB    U
ubuntu:latest               3131b4cc82a7        161MB         45.3MB    U
root@LAPTOP-HPUOTJF1:~# docker tag redis:latest achalpadol_redis:version1
root@LAPTOP-HPUOTJF1:~# docker images

IMAGE                       ID             DISK USAGE   CONTENT SIZE   EXTRA
achalpadol/ubuntu:latest    3131b4cc82a7        161MB         45.3MB    U
achalpadol/ubuntu:v1        3131b4cc82a7        161MB         45.3MB    U
achalpadol_redis:version1   234c902a2db4        208MB         56.3MB
httpd:latest                305fd8326a27        177MB         47.6MB    U
lancachenet/ubuntu:latest   f86a1b63836e        248MB         61.4MB
nginx:latest                b5a9a3cfc86b        241MB           66MB    U
redis:latest                234c902a2db4        208MB         56.3MB
ubuntu-nginx:latest         5102563991c6        315MB         80.4MB    U
ubuntu:latest               3131b4cc82a7        161MB         45.3MB    U
root@LAPTOP-HPUOTJF1:~# docker tag redis:latest achalpadol_redis:version2
root@LAPTOP-HPUOTJF1:~# docker tag redis:latest achalpadol_redis:version3
root@LAPTOP-HPUOTJF1:~# docker image s
docker: unknown command: docker image s

Usage:  docker image

Run 'docker image --help' for more information
root@LAPTOP-HPUOTJF1:~# docker images

IMAGE                       ID             DISK USAGE   CONTENT SIZE   EXTRA
achalpadol/ubuntu:latest    3131b4cc82a7        161MB         45.3MB    U
achalpadol/ubuntu:v1        3131b4cc82a7        161MB         45.3MB    U
achalpadol_redis:version1   234c902a2db4        208MB         56.3MB
achalpadol_redis:version2   234c902a2db4        208MB         56.3MB
achalpadol_redis:version3   234c902a2db4        208MB         56.3MB
```

```
root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# docker run -d --name redis01 achalpadol_redis:version1
b37b5a6ff268230f7a0f210832c2f78594b1e57e6763f014914d58cf6b5ef435
root@LAPTOP-HPUOTJF1:~# docker run -d --name redis02 achalpadol_redis:version3
ab6aaf3634f0dd3edd34088cc68201fe165010515b69fb6aadbf09a2cfcf1468
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE                       COMMAND                  CREATED          STATUS          PORTS                                         NAMES
ab6aaf3634f0   achalpadol_redis:version3   "docker-entrypoint.s…"   15 seconds ago   Up 14 seconds   6379/tcp                                      redis02
b37b5a6ff268   achalpadol_redis:version1   "docker-entrypoint.s…"   45 seconds ago   Up 44 seconds   6379/tcp                                      redis01
1b33c35e955b   nginx                       "/docker-entrypoint.…"   6 minutes ago    Up 6 minutes    0.0.0.0:80->80/tcp, [::]:80->80/tcp           nginx-web
977d5e5c948f   httpd                       "httpd-foreground"       28 minutes ago   Up 11 minutes   0.0.0.0:32768->80/tcp, [::]:32768->80/tcp     httpd-always
root@LAPTOP-HPUOTJF1:~# docker rmi achalpadol_redis:version2
Untagged: achalpadol_redis:version2
root@LAPTOP-HPUOTJF1:~# docker images
                                                                        i Info →   U In Use
IMAGE                       ID             DISK USAGE   CONTENT SIZE   EXTRA
achalpadol/ubuntu:latest    3131b4cc82a7        161MB         45.3MB    U
achalpadol/ubuntu:v1        3131b4cc82a7        161MB         45.3MB    U
achalpadol_redis:version1   234c902a2db4        208MB         56.3MB    U
achalpadol_redis:version3   234c902a2db4        208MB         56.3MB    U
httpd:latest                305fd8326a27        177MB         47.6MB    U
```

21. Clone the [GitHub - docker-library/httpd: Docker Official Image packaging for Apache HTTP Server](https://github.com/docker-library/httpd) Git repository locally and build the httpd Docker image from the Dockerfile at the "httpd/2.4/Dockerfile" path. Run and expose a Docker container from this image with a meaningful name on port 8181, then access the webpage in a browser. Afterward, tag the image as V1.

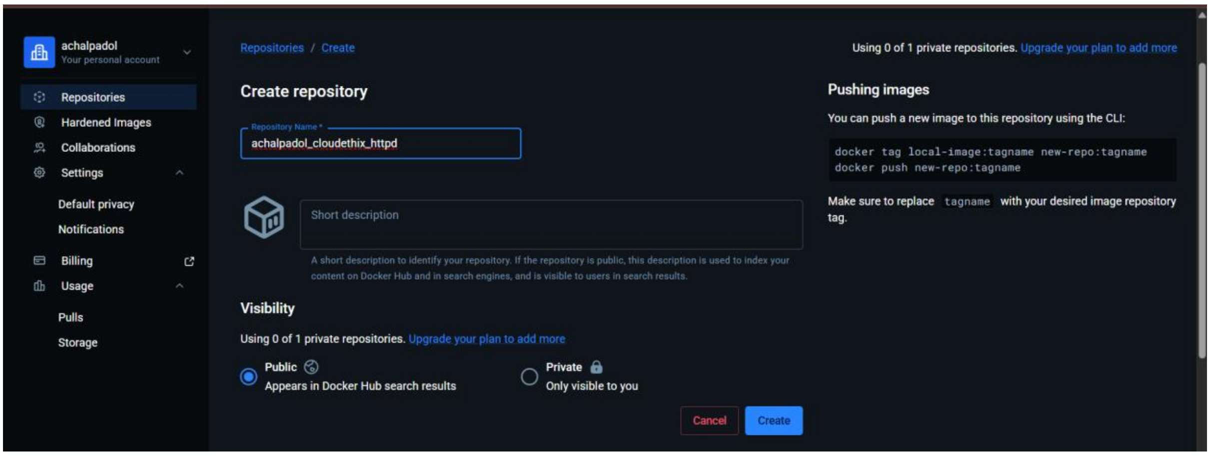
```
root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# git clone https://github.com/docker-library/httpd.git
Cloning into 'httpd'...
remote: Enumerating objects: 1119, done.
remote: Counting objects: 100% (205/205), done.
remote: Compressing objects: 100% (94/94), done.
remote: Total 1119 (delta 144), reused 117 (delta 111), pack-reused 914 (from 4)
Receiving objects: 100% (1119/1119), 227.61 KiB | 547.00 KiB/s, done.
Resolving deltas: 100% (487/487), done.
root@LAPTOP-HPUOTJF1:~# ls
1ff                   DevOps_Project  Merge_demo      clone-remote-repo     httpd                project       stash-demo    ubuntu-nginx
Amend_repo            GitIgnore-Repo  amend-repo      free-for-dev          index.html           revert-demo   stash1-demo
Cloud_Ethix_Copyfiles GitWorking      ci-cd-project   githubaction-project  merge_confilct_demo  revert_demo   student-repo
root@LAPTOP-HPUOTJF1:~# cd httpd/
root@LAPTOP-HPUOTJF1:~/httpd# ls
2.4  LICENSE  README.md  generate-stackbrew-library.sh  update.sh
root@LAPTOP-HPUOTJF1:~/httpd# docker build -t httpd-custom:latest -f 2.4/Dockerfile .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  388.6kB
Step 1/14 : FROM debian:trixie-slim
trixie-slim: Pulling from library/debian
8ff677ca96a9: Download complete
Digest: sha256:020c0d20b9880058cbe785a9db107156c3c75c2ac944a6aa7ab59f2add76a7bd
Status: Downloaded newer image for debian:trixie-slim
 ---> 020c0d20b988
Step 2/14 : ENV HTTPD_PREFIX /usr/local/apache2
 ---> Running in 93137df01e72
 ---> Removed intermediate container 93137df01e72
 ---> 2d237400f95a
Step 3/14 : ENV PATH $HTTPD_PREFIX/bin:$PATH
 ---> Running in 4564cb53df62
 ---> Removed intermediate container 4564cb53df62
 ---> d153592a9cd0
Step 4/14 : RUN mkdir -p "$HTTPD_PREFIX"        && chown www-data:www-data "$HTTPD_PREFIX"
 ---> Running in a0eee706afa9
 ---> Removed intermediate container a0eee706afa9
 ---> 99646c12e01b
Step 5/14 : WORKDIR $HTTPD_PREFIX
```

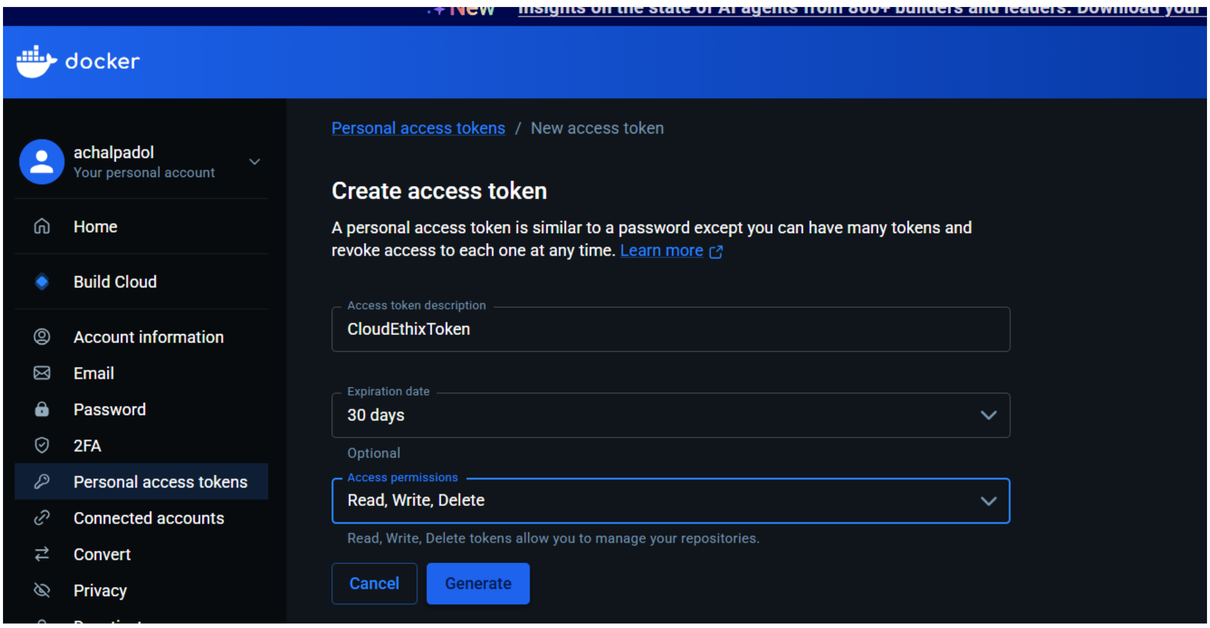
```
COPY failed: file not found in build context or excluded by .dockerignore: stat httpd-foreground: file does not exist
root@LAPTOP-HPUOTJF1:~/httpd# docker images
                                                                                              Info →  U In Use
IMAGE                       ID             DISK USAGE   CONTENT SIZE   EXTRA
achalpadol/ubuntu:latest    3131b4cc82a7        161MB         45.3MB    U
achalpadol/ubuntu:v1        3131b4cc82a7        161MB         45.3MB    U
achalpadol_redis:version1   234c902a2db4        208MB         56.3MB    U
achalpadol_redis:version3   234c902a2db4        208MB         56.3MB    U
debian:trixie-slim          020c0d20b988        119MB         31.8MB
httpd:latest                305fd8326a27        177MB         47.6MB    U
lancachenet/ubuntu:latest   f86a1b63836e        248MB         61.4MB
nginx:latest                b5a9a3cfc86b        241MB           66MB    U
redis:latest                234c902a2db4        208MB         56.3MB    U
ubuntu-nginx:latest         5102563991c6        315MB         80.4MB    U
ubuntu:latest               3131b4cc82a7        161MB         45.3MB    U
root@LAPTOP-HPUOTJF1:~/httpd# docker run -d --name apache-web -p 8181:80 httpd-custom:latest
Unable to find image 'httpd-custom:latest' locally
docker: Error response from daemon: pull access denied for httpd-custom, repository does not exist or may require 'docker login'

Run 'docker run --help' for more information
root@LAPTOP-HPUOTJF1:~/httpd# docker run -d --name apache-web -p 8181:80 httpd:latest
6ad5ea0758015fc2dad2bda5107ef2e807c01ffb0df45d6c6fca85c4d74b92ae
root@LAPTOP-HPUOTJF1:~/httpd# docker ps
CONTAINER ID   IMAGE          COMMAND              CREATED          STATUS          PORTS                                     NAMES
6ad5ea075801   httpd:latest   "httpd-foreground"   13 seconds ago   Up 12 seconds   0.0.0.0:8181->80/tcp, [::]:8181->80/tcp   apache-web
```

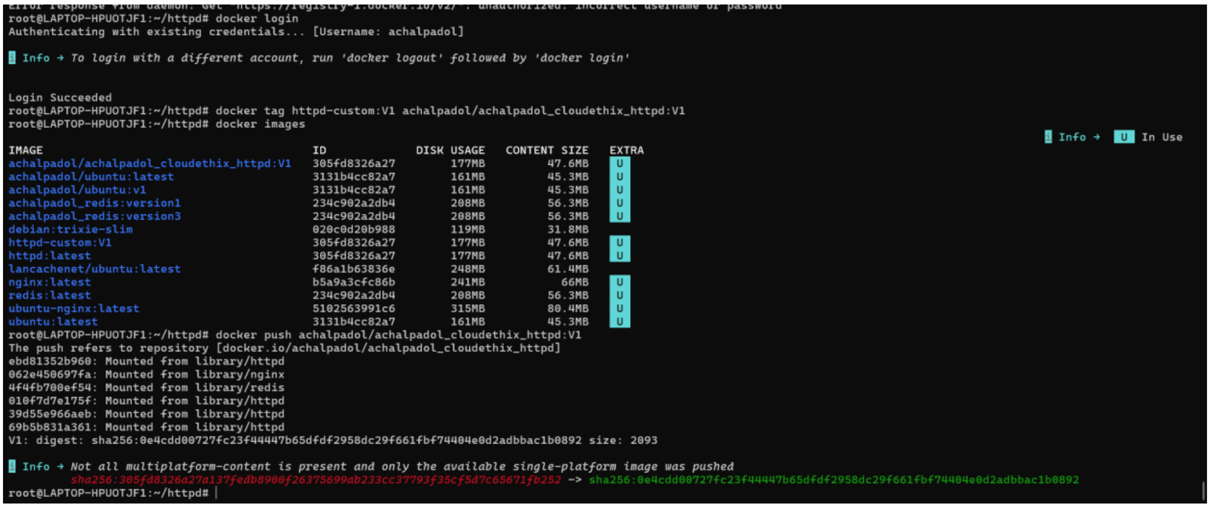
- tag the image as V1.

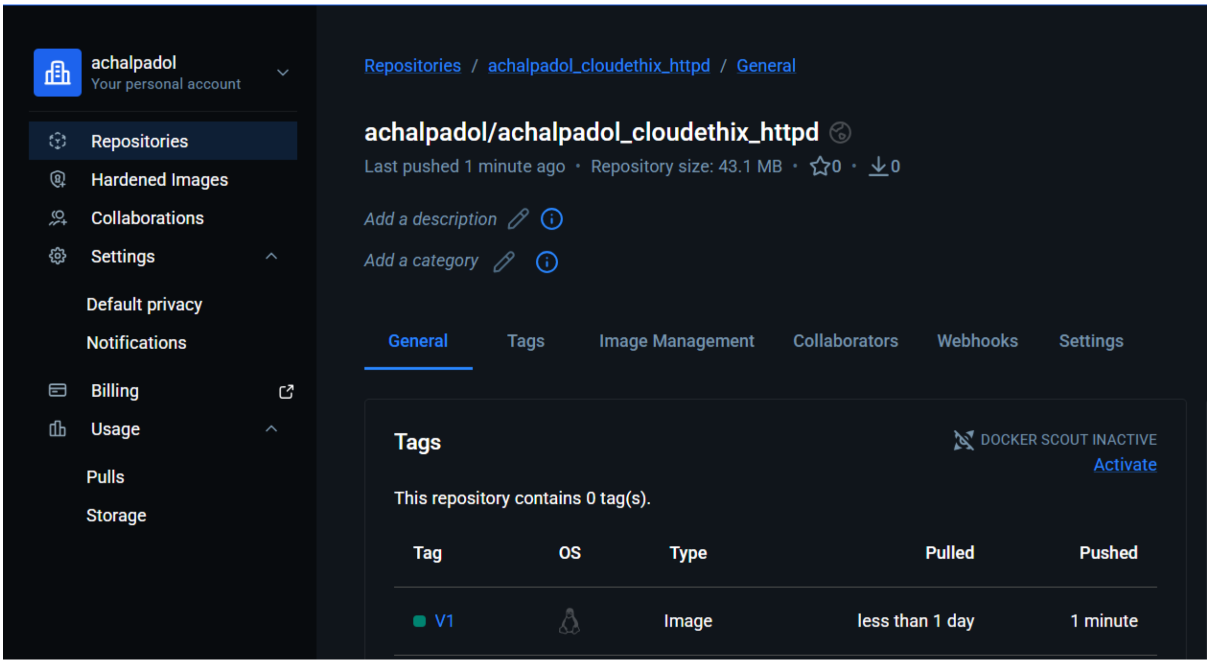
```
977d5e5c948f   httpd                       "httpd-foreground"       38 minutes ago
root@LAPTOP-HPUOTJF1:~/httpd# docker tag httpd:latest httpd-custom:V1
root@LAPTOP-HPUOTJF1:~/httpd# docker images

IMAGE                        ID             DISK USAGE   CONTENT SIZE   EXTRA
achalpadol/ubuntu:latest     3131b4cc82a7        161MB         45.3MB    U
achalpadol/ubuntu:v1         3131b4cc82a7        161MB         45.3MB    U
achalpadol_redis:version1    234c902a2db4        208MB         56.3MB    U
achalpadol_redis:version3    234c902a2db4        208MB         56.3MB    U
debian:trixie-slim           020c0d20b988        119MB         31.8MB
httpd-custom:V1              305fd8326a27        177MB         47.6MB    U
```

22. Create a Docker Hub account and generate a security token. Establish a Docker Hub repository named "yourname_cloudethix_httpd." Log in to Docker Hub and push the previously tagged httpd image to your repository.

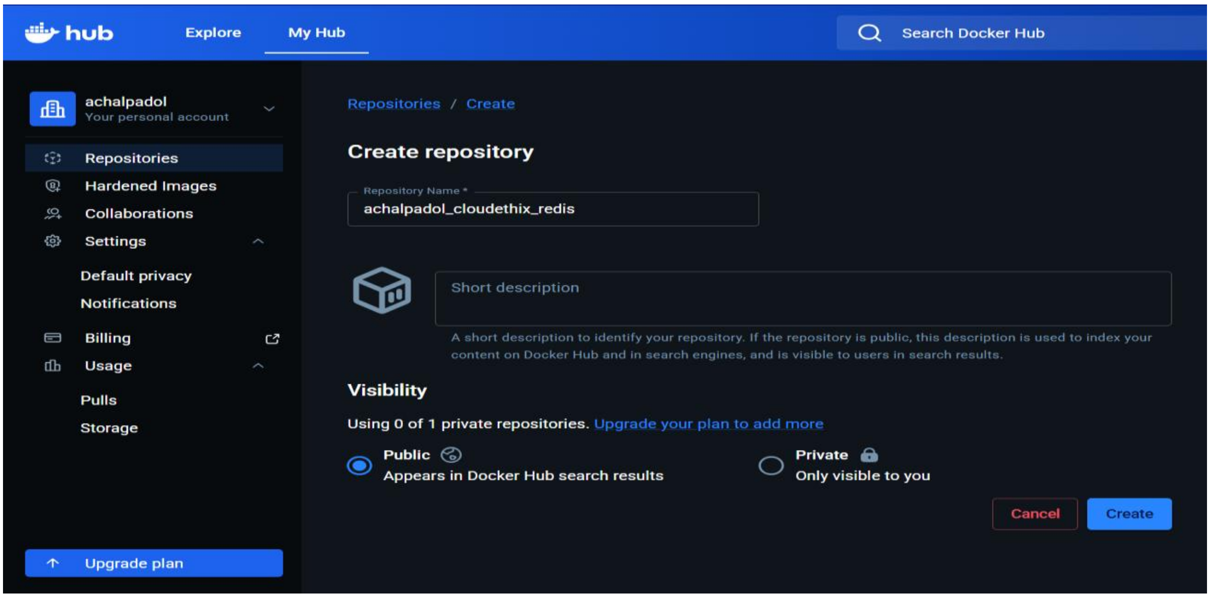


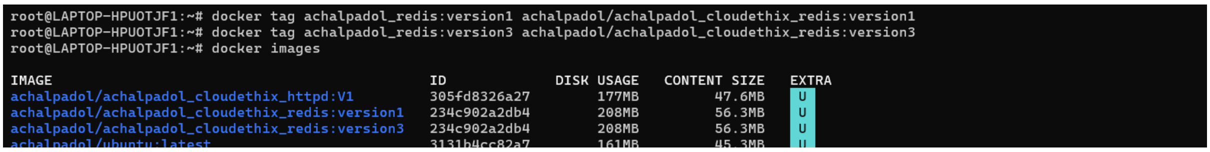


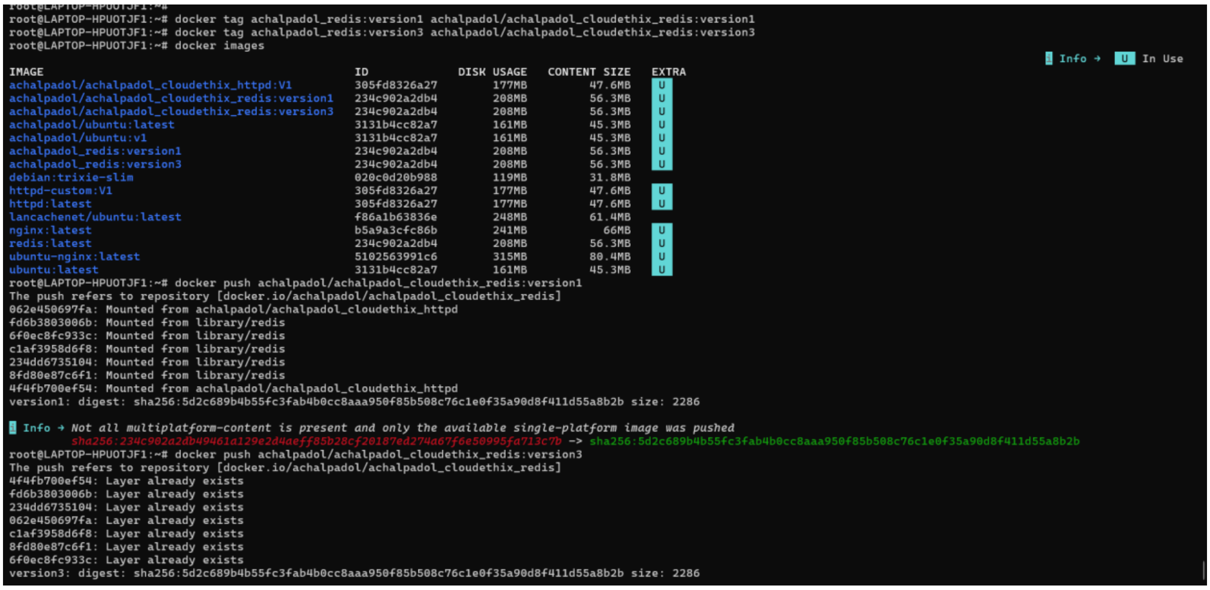


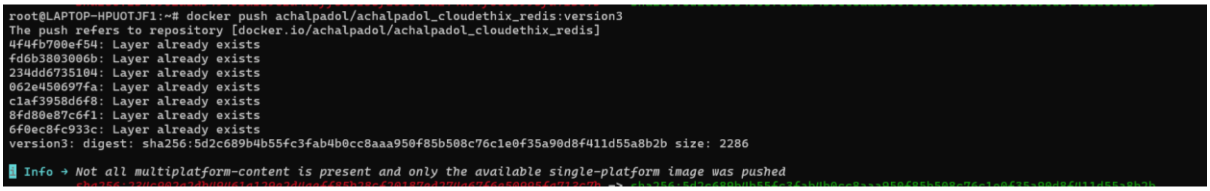


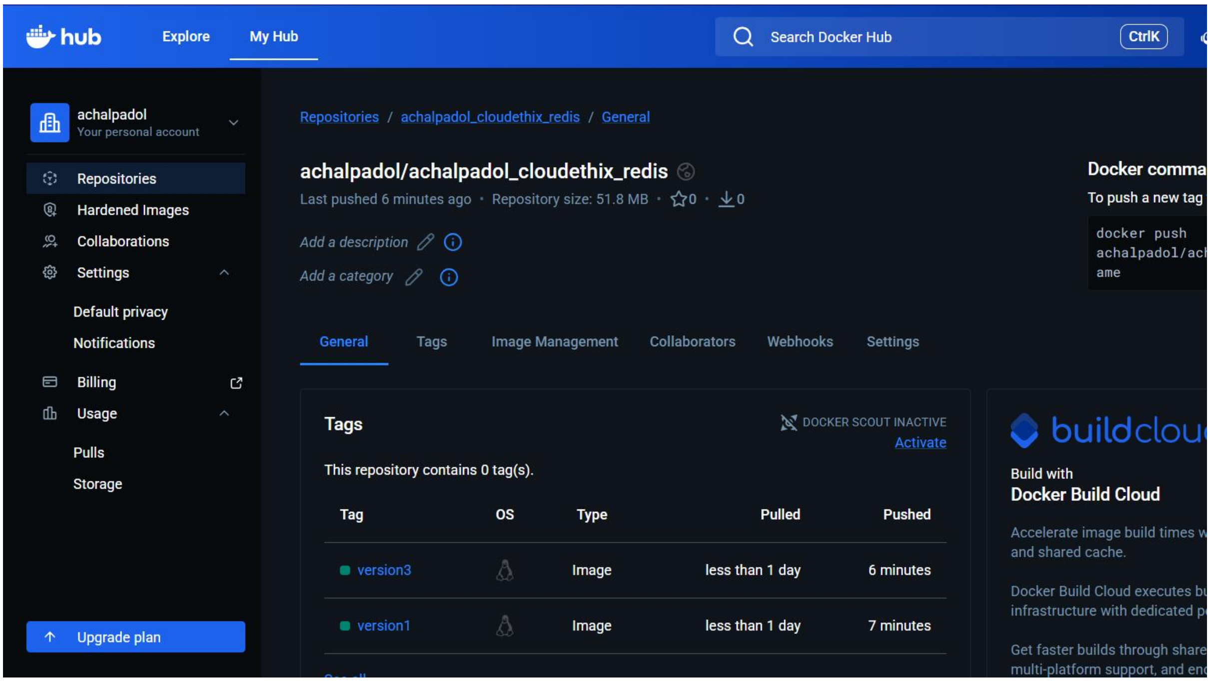
23. Push the Redis images tagged as version 1 and 3 to your Docker Hub repository, named "yourname_cloudethix_redis."











24. Create a remote Git repository and add the Dockerfile and index.html files. Clone the repository locally and create a release branch. Build the Docker image from the release branch with a meaningful tag, then run a container from the image and expose it on host port 8383. Check the webpage in a browser, and upon success, push the image to your

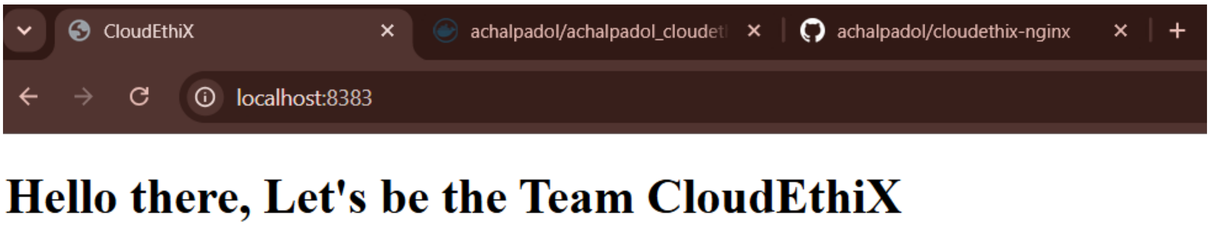
Docker Hub repository named "yourname_cloudethix_nginx." Finally, push the release branch to the remote Git repository and merge it by creating a pull request.

```
root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# git clone https://github.com/achalpadol/cloudethix-nginx.git
Cloning into 'cloudethix-nginx'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
root@LAPTOP-HPUOTJF1:~# ls
1ff                    DevOps_Project  Merge_demo     clone-remote-repo  githubaction-project  merge_confilct_demo  revert_demo  student-repo
Amend_repo             GitIgnore-Repo  amend-repo     cloudethix-nginx   httpd                 project              stash-demo   ubuntu-nginx
Cloud_Ethix_Copyfiles  GitWorking      ci-cd-project  free-for-dev       index.html            revert-demo          stash1-demo
root@LAPTOP-HPUOTJF1:~# cd cloudethix-nginx/
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# ls
Dockerfile  index.html
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# git checkout -b release
Switched to a new branch 'release'
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# git bracnh
git: 'bracnh' is not a git command. See 'git --help'.

The most similar command is
        branch
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# git branch
  main
* release
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker build -t nginx-release:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  66.56kB
Step 1/3 : FROM nginx:latest
 ---> b5a9a3cfc86b
Step 2/3 : COPY index.html /usr/share/nginx/html/index.html
 ---> cc17b8dbb198
Step 3/3 : EXPOSE 80
 ---> Running in efe48c186468
 ---> Removed intermediate container efe48c186468
 ---> e931316b8da7
Successfully built e931316b8da7
```

run a container from the image and expose it on host port 8383.

```
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker run -d --name nginx-release -p 8383:80 nginx-release:v1
7d6c8bf84dbbf319f7f8f57bd5a387a6d0c7444fad11397e737ba0f3e3bd5e0b
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker ps
CONTAINER ID   IMAGE              COMMAND                  CREATED          STATUS          PORTS                                     NAMES
7d6c8bf84dbb   nginx-release:v1   "/docker-entrypoint.…"   3 seconds ago    Up 3 seconds    0.0.0.0:8383->80/tcp, [::]:8383->80/tcp   nginx-release
6ad5ea075801   httpd:latest       "httpd-foreground"       44 minutes ago   Up 44 minutes   0.0.0.0:8181->80/tcp, [::]:8181->80/tcp   apache-web
```

. Check the webpage in a browser

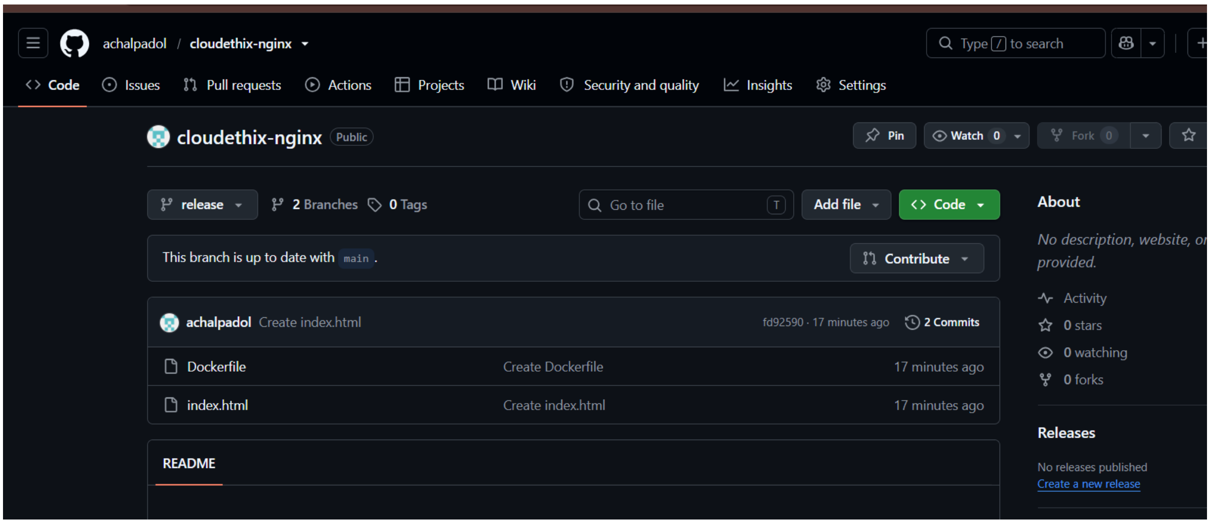


push the image to your Docker Hub repository named

```
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker push achalpadol/achalpadol_cloudethix_nginx:v1
The push refers to repository [docker.io/achalpadol/achalpadol_cloudethix_nginx]
39694cde734e: Mounted from library/nginx
45ef33eb91c8: Mounted from library/nginx
8d2092eabbc5: Mounted from library/nginx
ace9659054fc: Pushed
0b19be717994: Mounted from library/nginx
3b9f3509a826: Mounted from library/nginx
062e450697fa: Mounted from achalpadol/achalpadol_cloudethix_redis
6a24a6ebcf30: Mounted from library/nginx
v1: digest: sha256:e931316b8da71ccc4620e346cd5b3aabeb715870595105fe7547bb860041a30f size: 2043
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx#
```

Finally, push the release branch to the remote Git repository and merge it by creating a pull request.

```
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# git push -u origin release
Username for 'https://github.com': achalpadol
Password for 'https://achalpadol@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'release' on GitHub by visiting:
remote:      https://github.com/achalpadol/cloudethix-nginx/pull/new/release
remote:
To https://github.com/achalpadol/cloudethix-nginx.git
 * [new branch]      release -> release
branch 'release' set up to track 'origin/release'.
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# |
```



25. Save all local Redis images as a .tar file in the master branch of your local repository. Delete all Redis images from your local system and push the master branch to the remote repository. Load the Redis images from the tar file to your local system, and verify that all images have been loaded correctly.

```
Switched to branch 'master'
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker save -o redis-images.tar \
achalpadol_redis:version1 \
achalpadol_redis:version3
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# ls
Dockerfile  index.html  redis-images.tar
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# git add redis-images.tar
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# git commit -m "Added Redis Docker images tar file"
[master 8631763] Added Redis Docker images tar file
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 redis-images.tar
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# git push origin master
Username for 'https://github.com': achalpadol
Password for 'https://achalpadol@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 51.66 MiB | 914.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: warning: See https://gh.io/lfs for more information.
remote: warning: File redis-images.tar is 53.71 MB; this is larger than GitHub's recommended maximum file size of 50.00 MB
remote: warning: GH001: Large files detected. You may want to try Git Large File Storage - https://git-lfs.github.com.
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:      https://github.com/achalpadol/cloudethix-nginx/pull/new/master
remote:
To https://github.com/achalpadol/cloudethix-nginx.git
 * [new branch]      master -> master
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker rmi achalpadol_redis:version1
Untagged: achalpadol_redis:version1
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker rmi achalpadol_redis:version3
Untagged: achalpadol_redis:version3
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker images
                                                                            i Info →  U In Use
IMAGE                                      ID             DISK USAGE   CONTENT SIZE   EXTRA
achalpadol/achalpadol_cloudethix_httpd:V1  305fd8326a27        177MB         47.6MB    U
achalpadol/achalpadol_cloudethix_nginx:v1  e931316b8da7        238MB         63.1MB    U
```

. Load the Redis images from the tar file to your local system, and verify that all images have been loaded correctly.

```
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker load -i redis-images.tar
Loaded image: achalpadol_redis:version1
Loaded image: achalpadol_redis:version3
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker images
                                                                                 i Info →  U In Use
IMAGE                                            ID             DISK USAGE   CONTENT SIZE   EXTRA
achalpadol/achalpadol_cloudethix_httpd:V1        305fd8326a27        177MB         47.6MB    U
achalpadol/achalpadol_cloudethix_nginx:v1        e931316b8da7        238MB         63.1MB    U
achalpadol/achalpadol_cloudethix_redis:version1  234c902a2db4        208MB         56.3MB    U
achalpadol/achalpadol_cloudethix_redis:version3  234c902a2db4        208MB         56.3MB    U
achalpadol/ubuntu:latest                         3131b4cc82a7        161MB         45.3MB    U
achalpadol/ubuntu:v1                             3131b4cc82a7        161MB         45.3MB    U
achalpadol_redis:version1                        234c902a2db4        208MB         56.3MB    U
achalpadol_redis:version3                        234c902a2db4        208MB         56.3MB    U
debian:trixie-slim                               020c0d20b988        119MB         31.8MB
```

26. Pull the Busy-box image to your local system, tag it, and push it to the Docker Hub repository "yourname_cloudethix_busybox." Export the Docker image from the NGINX container, create a .tar file, and import the tar file to create a Docker image with a meaningful name. After importing the image, tag it and push it to the "yourname_cloudethix_busybox" Docker Hub repository.

- Pull the busy-box image to your local system

```
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# cd
root@LAPTOP-HPUOTJF1:~# docker pull busybox
Using default tag: latest
latest: Pulling from library/busybox
b05093807bb0: Pull complete
7270b3e1860c: Download complete
Digest: sha256:fd8d9aa63ba2f0982b5304e1ee8d3b90a210bc1ffb5314d980eb6962f1a9715d
Status: Downloaded newer image for busybox:latest
docker.io/library/busybox:latest
root@LAPTOP-HPUOTJF1:~# docker images

IMAGE                                          ID             DISK USAGE   CONTENT SIZE   EXTRA
achalpadol/achalpadol_cloudethix_httpd:V1      305fd8326a27        177MB         47.6MB    U
achalpadol/achalpadol_cloudethix_nginx:v1      e931316b8da7        238MB         63.1MB    U
achalpadol/achalpadol_cloudethix_redis:version1 234c902a2db4       208MB         56.3MB    U
achalpadol/achalpadol_cloudethix_redis:version3 234c902a2db4       208MB         56.3MB    U
achalpadol/ubuntu:latest                       3131b4cc82a7        161MB         45.3MB    U
achalpadol/ubuntu:v1                           3131b4cc82a7        161MB         45.3MB    U
achalpadol_redis:version1                      234c902a2db4        208MB         56.3MB    U
achalpadol_redis:version3                      234c902a2db4        208MB         56.3MB    U
busybox:latest                                 fd8d9aa63ba2       6.81MB         2.23MB
```

-tag it and push to the Docker Hub

```
ubuntu:latest                 3131b4cc82a7        161MB         45.3MB    U
root@LAPTOP-HPUOTJF1:~# docker push achalpadol/achalpadol_cloudethix_busybox:V1
The push refers to repository [docker.io/achalpadol/achalpadol_cloudethix_busybox]
b05093807bb0: Mounted from library/busybox
V1: digest: sha256:1cfa4e2b09e127b9c4ed43578d3f3c18e7d44ea47b9ea98475c0cbe9086525f8 size: 610

i Info → Not all multiplatform-content is present and only the available single-platform image was pushed
         sha256:fd8d9aa63ba2f0982b5304e1ee8d3b90a210bc1ffb5314d980eb6962f1a9715d -> sha256:1cfa4e2b09e127b9c4ed43578d3f3c18e7d44ea47b9ea98475c0cbe9086525f8
root@LAPTOP-HPUOTJF1:~# |
```

- Export the Docker image from the NGINX container, create a .tar file

```
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE                        COMMAND                  CREATED          STATUS          PORTS                                         NAMES
7d6c8bf84dbb   nginx-release:v1             "/docker-entrypoint.…"   46 minutes ago   Up 46 minutes   0.0.0.0:8383->80/tcp, [::]:8383->80/tcp       nginx-release
6ad5ea075801   httpd:latest                 "httpd-foreground"       2 hours ago      Up 2 hours      0.0.0.0:8181->80/tcp, [::]:8181->80/tcp       apache-web
ab6aaf3634f0   achalpadol_redis:version3    "docker-entrypoint.s…"   2 hours ago      Up 2 hours      6379/tcp                                      redis02
b37b5a6ff268   achalpadol_redis:version1    "docker-entrypoint.s…"   2 hours ago      Up 2 hours      6379/tcp                                      redis01
1b33c35e955b   nginx                        "/docker-entrypoint.…"   2 hours ago      Up 2 hours      0.0.0.0:80->80/tcp, [::]:80->80/tcp           nginx-web
977d5e5c948f   httpd                        "httpd-foreground"       2 hours ago      Up 2 hours      0.0.0.0:32768->80/tcp, [::]:32768->80/tcp     httpd-always
root@LAPTOP-HPUOTJF1:~# docker export -o nginx-container.tar nginx-web
root@LAPTOP-HPUOTJF1:~# ls
1ff                  DevOps_Project  Merge_demo      clone-remote-repo  githubaction-project  merge_conflict_demo  revert-demo  stash1-demo
Amend_repo           GitIgnore-Repo  amend-repo      cloudethix-nginx   httpd                 nginx-container.tar  revert_demo  student-repo
Cloud_Ethix_Copyfiles GitWorking     ci-cd-project   free-for-dev       index.html            project              stash-demo   ubuntu-nginx
```

- import the tar file to create a Docker image with a meaningful name.

```
Cloud_Ethix_Copyfiles GitWorking     ci-cd-project   free-for-dev       index.html            project              stash-demo   ubuntu-nginx
root@LAPTOP-HPUOTJF1:~# docker import nginx-container.tar nginx-imported:v1
sha256:ce7e04831e7acde10f06ba4a4807f6d9524e299121534a9999060c92f5826585
root@LAPTOP-HPUOTJF1:~# docker tag nginx-imported:v1 achalpadol/achalpadol_cloudethix_busybox:nginx-v1
root@LAPTOP-HPUOTJF1:~# docker push achalpadol/achalpadol_cloudethix_busybox:nginx-v1
The push refers to repository [docker.io/achalpadol/achalpadol_cloudethix_busybox]
23b11eaeb496: Pushed
nginx-v1: digest: sha256:ce7e04831e7acde10f06ba4a4807f6d9524e299121534a9999060c92f5826585 size: 406
root@LAPTOP-HPUOTJF1:~# docker images
                                                                                       i Info →  U In Use
IMAGE                                            ID             DISK USAGE   CONTENT SIZE   EXTRA
achalpadol/achalpadol_cloudethix_busybox:V1      fd8d9aa63ba2       6.81MB         2.23MB
achalpadol/achalpadol_cloudethix_busybox:nginx-v1 ce7e04831e7a      236MB         63.7MB
achalpadol/achalpadol_cloudethix_httpd:V1        305fd8326a27        177MB         47.6MB    U
achalpadol/achalpadol_cloudethix_nginx:v1        e931316b8da7        238MB         63.1MB    U
```

- push it to the "yourname_cloudethix_busybox" Docker Hub repository.

```
root@LAPTOP-HPUOTJF1:~# docker push achalpadol/achalpadol_cloudethix_busybox:V1
The push refers to repository [docker.io/achalpadol/achalpadol_cloudethix_busybox]
b05093807bb0: Mounted from library/busybox
V1: digest: sha256:1cfa4e2b09e127b9c4ed43578d3f3c18e7d44ea47b9ea98475c0cbe9086525f8 size: 610

i Info → Not all multiplatform-content is present and only the available single-platform image was pushed
         sha256:fd8d9aa63ba2f0982b5304e1ee8d3b90a210bc1ffb5314d980eb6962f1a9715d -> sha256:1cfa4e2b09e127b9c4ed43578d3f3c18e7d44ea47b9ea98475c0cbe9086525f8
root@LAPTOP-HPUOTJF1:~# docker ps
```

27. **Dockerfile creation**: Create a simple Dockerfile to build a custom image with the following specifications:
    - Base image: Ubuntu
    - Install packages: curl, vim, and git
    - Set an environment variable: MY_NAME=Your_Name

-Build the custom image using docker build and list all available images using docker images.

```
FROM ubuntu:latest

LABEL maintainer="Achal Padol"

RUN apt-get update && \
    apt-get install -y curl vim git && \
    apt-get clean && \
    rm -rf /var/lib/apt/lists/*

ENV MY_NAME="Achal Padol"

CMD ["/bin/bash"]
~
~
~
~
~
~
~
~
```

```
ubuntu:latest                                          3131b4cc82a7        161MB          45.3MB      U
root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# mkdir ubuntu-custom
root@LAPTOP-HPUOTJF1:~# cd ubuntu-custom/
root@LAPTOP-HPUOTJF1:~/ubuntu-custom# vim Dockerfile
root@LAPTOP-HPUOTJF1:~/ubuntu-custom# vim Dockerfile
root@LAPTOP-HPUOTJF1:~/ubuntu-custom# docker build -t ubuntu-custom:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.048kB
Step 1/5 : FROM ubuntu:latest
 ---> 3131b4cc82a7
Step 2/5 : LABEL maintainer="Achal Padol"
 ---> Running in f0301f9f134e
 ---> Removed intermediate container f0301f9f134e
 ---> 8ba7de75c849
Step 3/5 : RUN apt-get update &&     apt-get install -y curl vim git &&     apt-get clean &&     rm -rf /var/lib/apt/lists/*
```

```
Successfully built 5af056060ff5
Successfully tagged ubuntu-custom:v1
root@LAPTOP-HPUOTJF1:~/ubuntu-custom# docker images

IMAGE                                                ID             DISK USAGE   CONTENT SIZE   EXTRA
achalpadol/achalpadol_cloudethix_busybox:V1          fd8d9aa63ba2       6.81MB         2.23MB
achalpadol/achalpadol_cloudethix_busybox:nginx-v1    ce7e04831e7a        236MB         63.7MB
achalpadol/achalpadol_cloudethix_httpd:V1            305fd8326a27        177MB         47.6MB    U
achalpadol/achalpadol_cloudethix_nginx:v1            e931316b8da7        238MB         63.1MB    U
achalpadol/achalpadol_cloudethix_redis:version1      234c902a2db4        208MB         56.3MB    U
achalpadol/achalpadol_cloudethix_redis:version3      234c902a2db4        208MB         56.3MB    U
achalpadol/ubuntu:latest                             3131b4cc82a7        161MB         45.3MB    U
achalpadol/ubuntu:v1                                 3131b4cc82a7        161MB         45.3MB    U
achalpadol_redis:version1                            234c902a2db4        208MB         56.3MB    U
achalpadol_redis:version3                            234c902a2db4        208MB         56.3MB    U
busybox:latest                                       fd8d9aa63ba2       6.81MB         2.23MB
debian:trixie-slim                                   020c0d20b988        119MB         31.8MB
httpd-custom:V1                                      305fd8326a27        177MB         47.6MB    U
httpd:latest                                         305fd8326a27        177MB         47.6MB    U
lancachenet/ubuntu:latest                            f86a1b63836e        248MB         61.4MB
nginx-imported:v1                                    ce7e04831e7a        236MB         63.7MB
nginx-release:v1                                     e931316b8da7        238MB         63.1MB    U
nginx:latest                                         b5a9a3cfc86b        241MB           66MB    U
redis:latest                                         234c902a2db4        208MB         56.3MB    U
ubuntu-custom:v1                                     5af056060ff5        407MB         98.9MB
```

28. **Docker networking**: Create two Docker containers with different names, and add them to the same custom network. Verify that the containers can communicate with each other using their names as hostnames.

-Create Docker Network

```
root@LAPTOP-HPUOTJF1:~/ubuntu-custom# cd
root@LAPTOP-HPUOTJF1:~# docker network create cloud-network
a964394174fa54a200110e17d82484e7b10a72c6b8ffd3f98544bcea482fda69
root@LAPTOP-HPUOTJF1:~# docker network ls
NETWORK ID     NAME                          DRIVER    SCOPE
14516bb6eaf3   bridge                        bridge    local
a964394174fa   cloud-network                 bridge    local
```

-Create two Docker containers with different names, and add them to the same custom network.

```
root@LAPTOP-HPUOTJF1:~# docker run -dit --name ubuntu01 --network cloud-network ubuntu bash
9b93c1eae8897159c488110ed58ca63759afc484d1778ce3b6e960d180e2455e
root@LAPTOP-HPUOTJF1:~# docker run -dit --name ubuntu02 --network cloud-network ubuntu bash
d7f26af5dcf3797fdc721ea93335ff6c1a789778c5a557d62038a73605bb3d8b
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED          STATUS          PORTS     NAMES
d7f26af5dcf3   ubuntu    "bash"    5 seconds ago    Up 4 seconds              ubuntu02
9b93c1eae889   ubuntu    "bash"    19 seconds ago   Up 18 seconds             ubuntu01
```

```
root@9b93c1eae889:/# exit
exit
root@LAPTOP-HPUOTJF1:~# docker exec -it ubuntu01 bash
root@9b93c1eae889:/# ping ubuntu02
PING ubuntu02 (172.20.0.3) 56(84) bytes of data.
64 bytes from ubuntu02.cloud-network (172.20.0.3): icmp_seq=1 ttl=64 time=0.543 ms
64 bytes from ubuntu02.cloud-network (172.20.0.3): icmp_seq=2 ttl=64 time=0.302 ms
64 bytes from ubuntu02.cloud-network (172.20.0.3): icmp_seq=3 ttl=64 time=0.195 ms
64 bytes from ubuntu02.cloud-network (172.20.0.3): icmp_seq=4 ttl=64 time=0.132 ms
64 bytes from ubuntu02.cloud-network (172.20.0.3): icmp_seq=5 ttl=64 time=0.068 ms
```

29. **Data persistence with volumes**: Run a Docker container with an attached volume, create a file inside the volume, and verify that the file persists after the container is stopped and removed.

-Create a container with attach volume

```
root@9b93c1eae889:/# docker volume create cloud-data
bash: docker: command not found
root@9b93c1eae889:/# exit
exit
root@LAPTOP-HPUOTJF1:~# docker run -dit --name ubuntu-volume -v cloud-data:/data ubuntu bash
e43fbb4322545b75518db089111ded2abcfb86ec80f5dfd27050c6f5620d81a8
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED          STATUS          PORTS     NAMES
e43fbb432254   ubuntu    "bash"    14 seconds ago   Up 14 seconds             ubuntu-volume
d7f26af5dcf3   ubuntu    "bash"    11 minutes ago   Up 11 minutes             ubuntu02
```

-Create a file inside the volume

```
root@LAPTOP-HPUOTJF1:~# docker exec -it ubuntu-volume bash
root@e43fbb432254:/# echo "Hello CloudEthiX" > /data/file.txt
root@e43fbb432254:/# cat /data/file.txt
Hello CloudEthiX
root@e43fbb432254:/# exit
exit
root@LAPTOP-HPUOTJF1:~#
```

- verify that the file persists after the container is stopped and removed.

```
root@LAPTOP-HPUOTJF1:~# docker volume ls
DRIVER    VOLUME NAME
local     cab24f1de152a526001218ea642fd77196d83e8780175aff2e52fe311dd2696b
local     cloud-data
root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# docker rm -f ubuntu-volume
ubuntu-volume
root@LAPTOP-HPUOTJF1:~#
```

-create new container and attach volume to the new container and verify the data

```
root@LAPTOP-HPUOTJF1:~# docker rm -f ubuntu-volume
ubuntu-volume
root@LAPTOP-HPUOTJF1:~# docker run -dit --name ubuntu-volume-new \
-v cloud-data:/data \
ubuntu bash
c67f6cb39973ce71ec4f8801a54fc412fc040c18ec8927441036599afb4c3517
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED          STATUS          PORTS                                       NAMES
c67f6cb39973   ubuntu                         "bash"                   4 seconds ago    Up 3 seconds                                                ubuntu-volume-new
d7f26af5dcf3   ubuntu                         "bash"                   17 minutes ago   Up 17 minutes                                               ubuntu02
9b93c1eae889   ubuntu                         "bash"                   18 minutes ago   Up 18 minutes                                               ubuntu01
7d6c8bf84dbb   nginx-release:v1               "/docker-entrypoint.…"   2 hours ago      Up 2 hours      0.0.0.0:8383->80/tcp, [::]:8383->80/tcp     nginx-release
6ad5ea075801   httpd:latest                   "httpd-foreground"       2 hours ago      Up 2 hours      0.0.0.0:8181->80/tcp, [::]:8181->80/tcp     apache-web
ab6aaf3634f0   achalpadol_redis:version3      "docker-entrypoint.s…"   2 hours ago      Up 2 hours      6379/tcp                                    redis02
```

```
root@LAPTOP-HPUOTJF1:~#
docker exec -it ubuntu-volume-new bash
root@c67f6cb39973:/# ls /data
file.txt
root@c67f6cb39973:/# cat /data/file.txt
Hello CloudEthiX
root@c67f6cb39973:/#
```

30. **Docker Compose**: Create a docker-compose.yml file to run a multi-container application consisting of a web server (e.g., Nginx) and a database server (e.g., MySQL). Ensure that both containers are connected to the same network.

-create docker compose file to create multi-container

```yaml
version: "3.8"

services:
  web:
    image: nginx:latest
    container_name: nginx-web
    ports:
      - "8080:80"
    networks:
      - cloud-network

  db:
    image: mysql:8.0
    container_name: mysql-db
    environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: cloudethix
      MYSQL_USER: achal
      MYSQL_PASSWORD: password123
    ports:
      - "3306:3306"
    networks:
      - cloud-network

networks:
  cloud-network:
    driver: bridge
~
~
~
~
~
~
```

-run both the container

```
root@LAPTOP-HPUOTJF1:~/docker-compose-demo# ^C
root@LAPTOP-HPUOTJF1:~/docker-compose-demo# vim docker-compose.yml
root@LAPTOP-HPUOTJF1:~/docker-compose-demo# docker compose up -d
WARN[0000] /root/docker-compose-demo/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 2/2
 ✔Container nginx-web  Running                                                                                        0.0s
 ✔Container mysql-db   Started                                                                                        0.3s
root@LAPTOP-HPUOTJF1:~/docker-compose-demo# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                                  NAMES
cf35c688c237   mysql:8.0      "docker-entrypoint.s…"   12 seconds ago   Up 12 seconds   33060/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp   mysql-db
9e9d68ee8027   nginx:latest   "/docker-entrypoint.…"   3 minutes ago    Up 3 minutes    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp                nginx-web
c67f6cb39973   ubuntu         "bash"                   14 minutes ago   Up 14 minutes                                                          ubuntu-volum
e-new
```

- Ensure that both containers are connected to the same network.

```
        "com.docker.compose.project": "docker-compose-demo",
        "com.docker.compose.version": "2.40.3"
    },
    "Containers": {
        "9e9d68ee8027b762e315507a633f8dbf97fa6ee8c7257911f8aa6d6595e76649": {
            "Name": "nginx-web",
            "EndpointID": "17f1236ba6dc74bbda814b9aea25a5aa0ee31536490169522113bb9f152e62fb",
            "MacAddress": "7a:ce:29:2e:f5:de",
            "IPv4Address": "172.21.0.2/16",
            "IPv6Address": ""
        },
        "cf35c688c23712aa8bc4e8460b3a1a181b5d698e4cacdbc16d08283f07dcb174": {
            "Name": "mysql-db",
            "EndpointID": "2c92904f3790cef5da502e1e7a6f429aada6d2b778560dc3ea4cb0c78c076e61",
            "MacAddress": "5e:1f:09:5c:64:71",
            "IPv4Address": "172.21.0.3/16",
            "IPv6Address": ""
        }
```

31. **Container logging**: Run a Docker container that generates log output, and practice using docker logs to view and analyze the logs.

```
root@LAPTOP-HPUOTJF1:~# docker run -d --name nginx-logs -p 8081:80 nginx
8f30dbddbbf1271508b55fdfaf6345de056fc37e810ad59709168cca9a8bdbaf
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                     NAMES
8f30dbddbbf1   nginx     "/docker-entrypoint.…"   6 seconds ago   Up 5 seconds   0.0.0.0:8081->80/tcp, [::]:8081->80/tcp   nginx-logs
```

View logs

```
root@LAPTOP-HPUOTJF1:~# docker logs nginx-logs
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/22 06:40:26 [notice] 1#1: using the "epoll" event method
2026/07/22 06:40:26 [notice] 1#1: nginx/1.31.2
2026/07/22 06:40:26 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/22 06:40:26 [notice] 1#1: OS: Linux 6.6.114.1-microsoft-standard-WSL2
2026/07/22 06:40:26 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
2026/07/22 06:40:26 [notice] 1#1: start worker processes
2026/07/22 06:40:26 [notice] 1#1: start worker process 29
2026/07/22 06:40:26 [notice] 1#1: start worker process 30
2026/07/22 06:40:26 [notice] 1#1: start worker process 31
2026/07/22 06:40:26 [notice] 1#1: start worker process 32
2026/07/22 06:40:26 [notice] 1#1: start worker process 33
2026/07/22 06:40:26 [notice] 1#1: start worker process 34
2026/07/22 06:40:26 [notice] 1#1: start worker process 35
2026/07/22 06:40:26 [notice] 1#1: start worker process 36
2026/07/22 06:40:26 [notice] 1#1: start worker process 37
2026/07/22 06:40:26 [notice] 1#1: start worker process 38
2026/07/22 06:40:26 [notice] 1#1: start worker process 39
2026/07/22 06:40:26 [notice] 1#1: start worker process 40
172.17.0.1 - - [22/Jul/2026:06:40:55 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:40:59 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:01 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:02 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:03 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:05 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
root@LAPTOP-HPUOTJF1:~#
```

```
^Croot@LAPTOP-HPUOTJF1:~docker logs --tail 5 nginx-logsgs
172.17.0.1 - - [22/Jul/2026:06:40:59 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:01 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:02 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:03 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:05 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
root@LAPTOP-HPUOTJF1:~#
```

32. Container resource monitoring: Run a Docker container that consumes system resources (e.g., CPU or memory) and practice using docker stats and docker top to monitor the container's resource usage.

-docker stats

```
root@LAPTOP-HPUOTJF1:~#
CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT     MEM %     NET I/O           BLOCK I/O         PIDS
7a0dc5581c86   cpu-demo            0.00%     948KiB / 7.581GiB     0.01%     796B / 126B       7.66MB / 4.1kB    1
8f30dbddbbf1   nginx-logs          0.00%     10.54MiB / 7.581GiB   0.14%     4.39kB / 9kB      3.43MB / 12.3kB   13
cf35c688c237   mysql-db            1.43%     385.8MiB / 7.581GiB   4.97%     1.01kB / 126B     72MB / 266MB      37
9e9d68ee8027   nginx-web           0.00%     10.46MiB / 7.581GiB   0.13%     1.7kB / 126B      3.42MB / 12.3kB   13
c67f6cb39973   ubuntu-volume-new   0.00%     956KiB / 7.581GiB     0.01%     1.37kB / 126B     487kB / 4.1kB     1
d7f26af5dcf3   ubuntu02            0.00%     936KiB / 7.581GiB     0.01%     4.65kB / 3.58kB   0B / 0B           1
9b93c1eae889   ubuntu01            0.00%     5.406MiB / 7.581GiB   0.07%     26.5MB / 439kB    40.7MB / 69.7MB   1
```

-docker stat cpu-demo

```
CONTAINER ID   NAME       CPU %     MEM USAGE / LIMIT   MEM %     NET I/O       BLOCK I/O        PIDS
7a0dc5581c86   cpu-demo   0.00%     948KiB / 7.581GiB   0.01%     866B / 126B   7.66MB / 4.1kB   1
```

-docker top cup-demo

```
^Croot@LAPTOP-HPUOTJF1:~# docker top cpu-demo
UID     PID     PPID    C    STIME    TTY     TIME       CMD
root    68914   68890   0    06:47    pts/0   00:00:00   bash
root@LAPTOP-HPUOTJF1:~#
```

33. **Updating a containerized application**: Update the source code of a simple containerized application, rebuild the Docker image, and deploy the updated version while minimizing downtime.

-index.html file

```
<!DOCTYPE html>
<html>
<head>
    <title>CloudEthiX</title>
</head>
<body>
    <h1>Version 1</h1>
    <h2>Hello from CloudEthiX</h2>
</body>
</html>
~
~
~
~
~
```

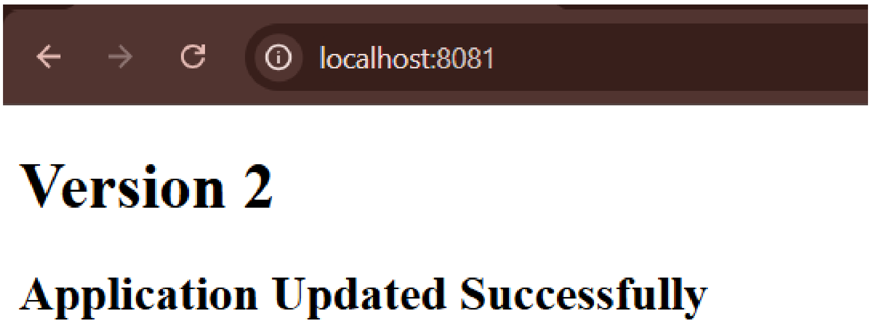
-build docker image form the created Dockerfile

```
root@LAPTOP-HPUOTJF1:~/nginx-update# docker build -t nginx-app:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/3 : FROM nginx:latest
 ---> b5a9a3cfc86b
Step 2/3 : COPY index.html /usr/share/nginx/html/index.html
 ---> 52aadc059802
Step 3/3 : EXPOSE 80
 ---> Running in 36f8efd2c144
 ---> Removed intermediate container 36f8efd2c144
 ---> 9d124c9d5add
Successfully built 9d124c9d5add
Successfully tagged nginx-app:v1
root@LAPTOP-HPUOTJF1:~/nginx-update# docker images

IMAGE                                                ID             DISK USAGE   CONTENT SIZE   EXTRA
achalpadol/achalpadol_cloudethix_busybox:V1          fd8d9aa63ba2       6.81MB         2.23MB
achalpadol/achalpadol_cloudethix_busybox:nginx-v1    ce7e04831e7a        236MB         63.7MB
achalpadol/achalpadol_cloudethix_httpd:V1            305fd8326a27        177MB         47.6MB    U
achalpadol/achalpadol_cloudethix_nginx:v1            e931316b8da7        238MB         63.1MB    U
achalpadol/achalpadol_cloudethix_redis:version1      234c902a2db4        208MB         56.3MB    U
achalpadol/achalpadol_cloudethix_redis:version3      234c902a2db4        208MB         56.3MB    U
achalpadol/ubuntu:latest                             3131b4cc82a7        161MB         45.3MB    U
achalpadol/ubuntu:v1                                 3131b4cc82a7        161MB         45.3MB    U
achalpadol_redis:version1                            234c902a2db4        208MB         56.3MB    U
achalpadol_redis:version3                            234c902a2db4        208MB         56.3MB    U
busybox:latest                                       fd8d9aa63ba2       6.81MB         2.23MB
debian:trixie-slim                                   020c0d20b988        119MB         31.8MB
httpd-custom:V1                                      305fd8326a27        177MB         47.6MB    U
httpd:latest                                         305fd8326a27        177MB         47.6MB    U
lancachenet/ubuntu:latest                            f86a1b63836e        248MB         61.4MB
mysql:8.0                                            7dcddc01f13b        1.1GB          249MB    U
nginx-app:v1                                         9d124c9d5add        238MB         63.1MB
```

-Update the source code file

```
<!DOCTYPE html>
<html>
<head>
    <title>CloudEthiX</title>
</head>
<body>
    <h1>Version 2</h1>
    <h2>Application Updated Successfully</h2>
</body>
</html>
~
~
~
~
~
~
~
~
~
~
```

-access the application through the browser



34. **Bridge network**: Create two Docker containers using different images, such as NGINX and MySQL. Connect them using a user-defined bridge network and ensure they can communicate with each other.

-create two docker containers using different images such as NGINX and MySQL

```
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE       COMMAND                  CREATED          STATUS          PORTS                                   NAMES
d2e110a3dd5d   nginx       "/docker-entrypoint.…"   3 minutes ago    Up 3 minutes    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   nginx-web
d6f0c28eb95a   mysql:8.0   "docker-entrypoint.s…"   13 minutes ago   Up 13 minutes   3306/tcp, 33060/tcp                     mysql-db
root@LAPTOP-HPUOTJF1:~# docker images

IMAGE          ID             DISK USAGE   CONTENT SIZE   EXTRA
mysql:8.0      7dcddc01f13b        1.1GB          249MB    U
nginx:latest   5a88c9c45479        241MB           66MB    U
root@LAPTOP-HPUOTJF1:~# |
```

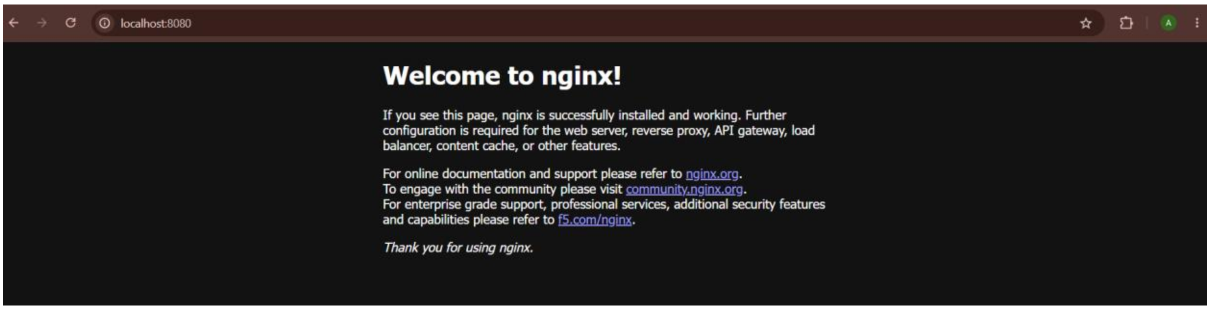
- ensure they can communicate with each other.

```
        "ConfigOnly": false,
        "Options": {},
        "Labels": {},
        "Containers": {
            "d2e110a3dd5d23c9c7b97ec5606908e18838abe343fd22429196eba42be35db0": {
                "Name": "nginx-web",
                "EndpointID": "0fa7c102cc595cdf285434281163a2d2e4ddc0d7c4a198de67912064980dd30e",
                "MacAddress": "a6:7a:99:ff:0b:21",
                "IPv4Address": "172.20.0.3/16",
                "IPv6Address": ""
            },
            "d6f0c28eb95ae53afc0e1b98dd44e6a2505873d41bac4f3fe6d0f11d72a0293a": {
                "Name": "mysql-db",
                "EndpointID": "5b8a411c950b55ee85d63927df4e23561d09ab46638b7cd4679d69d7a6ddfba8",
                "MacAddress": "f2:41:c3:92:f5:27",
                "IPv4Address": "172.20.0.2/16",
                "IPv6Address": ""
            }
        },
        "Status": {
```

35. **Port mapping**: Run a containerized web server (e.g., NGINX or Apache) and map its default port (80 or 443) to a custom port on the host machine. Verify that the web server is accessible through the custom port on the host.

```
root@LAPTOP-HPUOTJF1:~# docker run -d --name nginx-port -p 8080:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
d26f27cc8c41: Pull complete
cacfcdd01f30: Pull complete
3c7ab7949321: Pull complete
b6698f04e005: Pull complete
2bedaf25031a: Pull complete
82454cdbf456: Pull complete
ea1d76ccc2c6: Download complete
6c496f5b5050: Download complete
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Downloaded newer image for nginx:latest
b92d613b0a34d9df8aeccd3d398279cecf3d73abeff2754da1f2e5605e1bedb9
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE       COMMAND                  CREATED          STATUS          PORTS                                   NAMES
b92d613b0a34   nginx       "/docker-entrypoint.…"   6 seconds ago    Up 5 seconds    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   nginx-port
d6f0c28eb95a   mysql:8.0   "docker-entrypoint.s…"   21 minutes ago   Up 21 minutes   3306/tcp, 33060/tcp                     mysql-db
root@LAPTOP-HPUOTJF1:~# |
```

- Verify that the web server is accessible through the custom port on the host.




```
d6f0c28eb95a   mysql:8.0   "docker-entrypoint.s…"
root@LAPTOP-HPUOTJF1:~# docker port nginx-port
80/tcp -> 0.0.0.0:8080
80/tcp -> [::]:8080
root@LAPTOP-HPUOTJF1:~# docker inspect nginx-port
```

```
            "StopSignal": "SIGQUIT"
        },
        "NetworkSettings": {
            "SandboxID": "a19966f277b01ded172f99cb1aea641410945c8fdf1aeba577649965d2c2ff8a",
            "SandboxKey": "/var/run/docker/netns/a19966f277b0",
            "Ports": {
                "80/tcp": [
                    {
                        "HostIp": "0.0.0.0",
                        "HostPort": "8080"
                    },
                    {
                        "HostIp": "::",
                        "HostPort": "8080"
                    }
                ]
```

36. **Host networking**: Run a container with the host networking mode and compare its network configuration with that of the host machine. Explain the advantages and disadvantages of using host networking for containers.

- Run a container with the host networking mode and compare its network configuration

```
root@LAPTOP-HPUOTJF1:~# docker run -d --name nginx-host --network host nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
d26f27cc8c41: Pull complete
2bedaf25031a: Pull complete
b6698f04e005: Pull complete
cacfcdd01f30: Pull complete
3c7ab7949321: Pull complete
82454cdbf456: Pull complete
6c496f5b5050: Download complete
ea1d76ccc2c6: Download complete
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Downloaded newer image for nginx:latest
44c64692a9a63fe9c9563fe5b5fa6c62bffb9a396d6f3fd2d51190affe611556
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS     NAMES
44c64692a9a6   nginx     "/docker-entrypoint.…"   4 seconds ago   Up 3 seconds             nginx-host
root@LAPTOP-HPUOTJF1:~# hostname -I
172.18.178.20 172.17.0.1 172.19.0.1 172.20.0.1 172.21.0.1 172.22.0.1
root@LAPTOP-HPUOTJF1:~# |
```

- Check the container's network configuration

```
exit
root@LAPTOP-HPUOTJF1:~# docker exec -it nginx-host /bin/bash
root@LAPTOP-HPUOTJF1:/# hostname -I
172.18.178.20 172.17.0.1 172.19.0.1 172.20.0.1 172.21.0.1 172.22.0.1
root@LAPTOP-HPUOTJF1:/# |
```

Advantages of Host Networking

1. **Better Performance**
   - No Network Address Translation (NAT) is used.
   - Packets travel directly through the host's network stack, resulting in lower latency and higher throughput.
2. **No Port Mapping Required**
   - You don't need to use the -p or --publish option.
   - Applications running in the container are directly accessible through the host's network.
3. **Direct Access to Host Network**
   - The container can use the host's network interfaces and IP address.
   - Useful for applications that need direct access to the network.
4. **Suitable for High-Performance Applications**
   - Commonly used for network monitoring tools, proxies, load balancers, and applications where network performance is critical.

Disadvantages of Host Networking

1. **No Network Isolation**
   - The container shares the host's network namespace.
   - It does not have its own separate IP address.
2. **Port Conflicts**
   - If the host is already using a port (for example, port 80), another container using host networking cannot use the same port.
   - Only one application can bind to a given port on the host.
3. **Reduced Security**
   - Since the container shares the host's network, it has greater access to the host's networking resources.
   - This increases the potential impact if the container is compromised.
4. **Less Flexible**
   - Host networking is not ideal when running multiple instances of the same application on the same machine because they cannot listen on the same ports.

37. **Named volume**: Launch a containerized database (e.g., MySQL or PostgreSQL) using a named volume to store its data. Stop and remove the container, then recreate it using the same named volume. Verify that the data persists across container recreations.

-Launch a containerized database using named volume to store its data

```
root@LAPTOP-HPUOTJF1:~# docker run -d --name mysql-db -v mysql-data:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=root123 -e MYSQL_DATABASE=cloudethix -p 3307:3306 mysql:8.0
Unable to find image 'mysql:8.0' locally
8.0: Pulling from library/mysql
4c8a3e0d4e4b: Pull complete
6ef6c7b50a93: Pull complete
a63160a5eda1: Pull complete
0d74d296605b: Pull complete
e3e5d1ac74c1: Pull complete
edf85873f64e: Pull complete
297d04cfe470: Pull complete
7534d1db9f8d: Pull complete
49ec2dab01d9: Pull complete
ab24264a27e9: Pull complete
96d30d9fbee8: Pull complete
796812c73292: Download complete
af166387641d: Download complete
Digest: sha256:7dcddc01f13bab2f15cde676d44d01f61fc9f99fe7785e86196dfc07d358ae2b
Status: Downloaded newer image for mysql:8.0
d302b015ae88de35ffcf384e9e6c62f722d9b6d1a84d8957c021c26835bd4de9
root@LAPTOP-HPUOTJF1:~# docker volume ls
DRIVER    VOLUME NAME
local     5b5beeb723e2e7d4009004e10742c1926190b05cb52bf2ac6dc6b579aa743ce2
local     cab24f1de152a526001218ea642fd77196d83e8780175aff2e52fe311dd2696b
local     cloud-data
local     f4a69a7ecc58d3de714aeac3016399c1ab5d3f01f4e66a031830bf8e8f92fcf0
local     mysql-data
```

```
local     mysql-data
root@LAPTOP-HPUOTJF1:~# docker exec -it mysql-db mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE cloudethix;
Database changed
mysql> CREATE TABLE employee (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO employee VALUES (1,'Achal');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM employee;
+----+-------+
| id | name  |
+----+-------+
|  1 | Achal |
+----+-------+
1 row in set (0.00 sec)
```

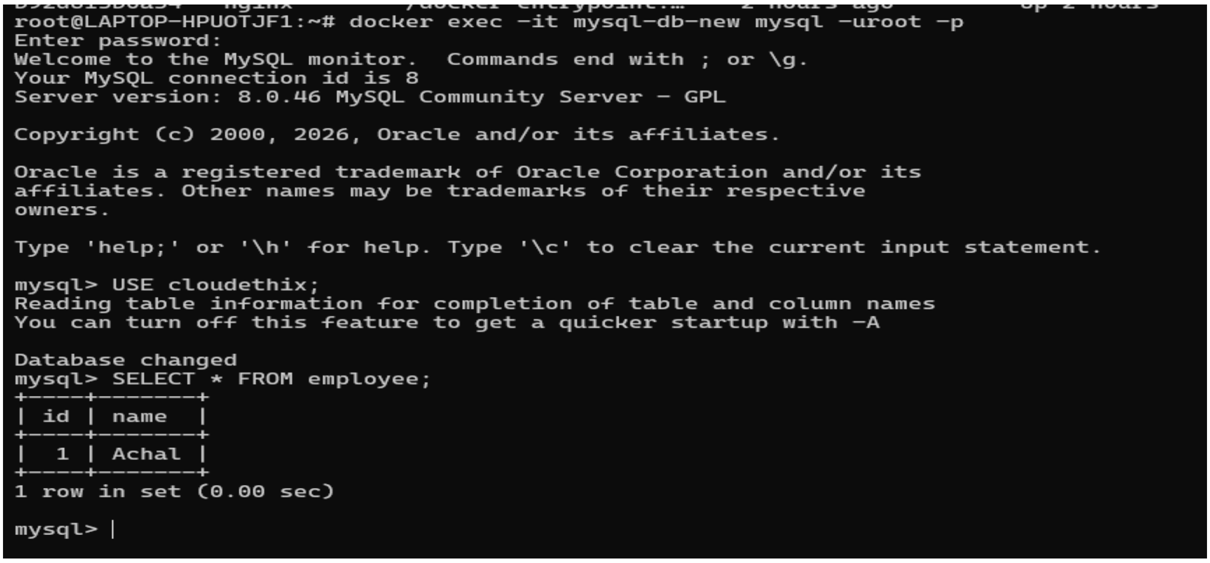
-stop and remove the container

```
Bye
root@LAPTOP-HPUOTJF1:~# docker stop mysql-db
mysql-db
root@LAPTOP-HPUOTJF1:~# docker rm mysql-db
mysql-db
root@LAPTOP-HPUOTJF1:~# docker run -d --name mysql-db-new -v mysql-data:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=root123 -e MYSQL_DATABASE=cloudethix -p 3307:3306 mysql:8.0
8f03e78845095241703bd56e49c7265cc06af14677b9c7615a4c8bcee8614c6d
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE       COMMAND                  CREATED              STATUS              PORTS                                                    NAMES
8f03e7884509   mysql:8.0   "docker-entrypoint.s…"   6 seconds ago        Up 6 seconds        33060/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp   mysql-db-new
467f1cae5567   nginx       "/docker-entrypoint.…"   About an hour ago    Up About an hour                                                             nginx-host
b92d613b0a34   nginx       "/docker-entrypoint.…"   2 hours ago          Up 2 hours          0.0.0.0:8080->80/tcp, [::]:8080->80/tcp                  nginx-port
root@LAPTOP-HPUOTJF1:~# |
```

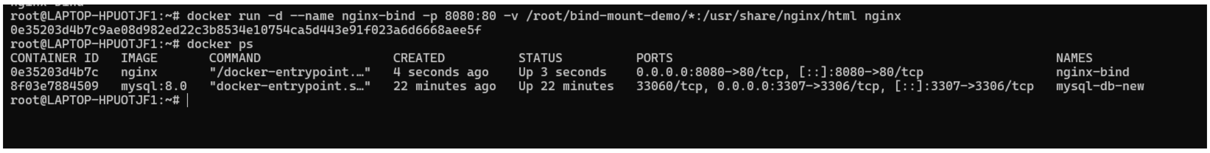
-recreate it using a same name volume

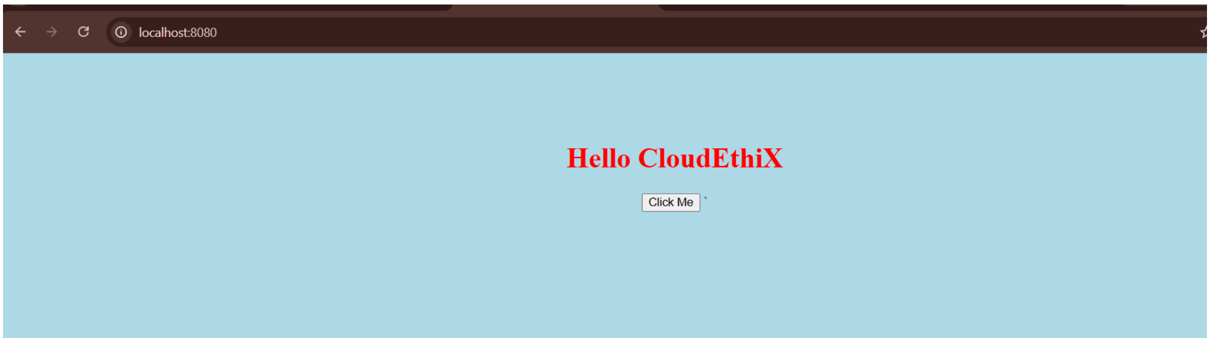
```
mysql-db
root@LAPTOP-HPUOTJF1:~# docker run -d --name mysql-db-new -v mysql-data:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=root123 -e MYSQL_DATABASE=cloudethix -p 3307:3306 mysql:8.0
8f03e78845095241703bd56e49c7265cc06af14677b9c7615a4c8bcee8614c6d
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE       COMMAND                  CREATED              STATUS              PORTS                                                    NAMES
8f03e7884509   mysql:8.0   "docker-entrypoint.s…"   6 seconds ago        Up 6 seconds        33060/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp   mysql-db-new
467f1cae5567   nginx       "/docker-entrypoint.…"   About an hour ago    Up About an hour                                                             nginx-host
b92d613b0a34   nginx       "/docker-entrypoint.…"   2 hours ago          Up 2 hours          0.0.0.0:8080->80/tcp, [::]:8080->80/tcp                  nginx-port
root@LAPTOP-HPUOTJF1:~# |
```

- Verify that the data persists across container recreations.



38. **Bind mount**: Create a simple web application with a local directory containing HTML, CSS, and JavaScript files. Run a web server container (e.g., NGINX or Apache) and bind mount the local directory to the container's webroot. Verify that the web application is accessible and update the local files to see if changes are reflected in the container.





-update index.html file local file

```
<!DOCTYPE html>
<html>
<head>
    <title>CloudEthiX</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <h1>Bind Mount Working Successfully</h1>
    <p>Updated without rebuilding the Docker image.</p>

    <button onclick="showMessage()">
        Click Me
    </button>

    <script src="script.js"></script>
</body>
</html>`
~
~
~
~
~
~
```

- Verify that the web application is accessible and update the local files to see if changes are reflected in the container.

```
root@LAPTOP-HPUOTJF1:~/bind-mount-demo# vim index.html
root@LAPTOP-HPUOTJF1:~/bind-mount-demo# docker exec -it nginx-bind cat /usr/share/nginx/html/index.html
<!DOCTYPE html>
<html>
<head>
    <title>CloudEthiX</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <h1>Bind Mount Working Successfully</h1>
    <p>Updated without rebuilding the Docker image.</p>

    <button onclick="showMessage()">
        Click Me
    </button>

    <script src="script.js"></script>
</body>
</html>`
root@LAPTOP-HPUOTJF1:~/bind-mount-demo# |
```

39. **Volume backups**: Using a container with a named volume, create a backup of the volume's data by either exporting it as a tarball or copying the data to a host directory. Restore the data to a new container and verify that the restoration was successful.

-Create a container using a name volume

```
root@LAPTOP-HPUOTJF1:~# docker run -d --name mysql-db -v mysql-data:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=root123 -e MYSQL_DATABASE=company -p 3307:3306 mysql:8.0
c5cd604005937a93734852221dbf17ace8f9cf4b9e51efe0e33ef1d5ac11d88e
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE       COMMAND                  CREATED          STATUS          PORTS                                                      NAMES
c5cd60400593   mysql:8.0   "docker-entrypoint.s…"   41 seconds ago   Up 40 seconds   33060/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp   mysql-db
012bbee786a4   nginx       "/docker-entrypoint.…"   40 minutes ago   Up 40 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp                    nginx-bind
root@LAPTOP-HPUOTJF1:~# docker exec -it mysql-db mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE company;
EATE TABLE employee(
id INT,
name VARCHAR(50)
);

INSERT INTO employee VALUES(1,'Achal');

SELECT * FROM employee;Database changed
mysql>
mysql> CREATE TABLE employee(
    -> id INT,
    -> name VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> INSERT INTO employee VALUES(1,'Achal');
Query OK, 1 row affected (0.02 sec)

mysql>
mysql> SELECT * FROM employee;
+------+-------+
| id   | name  |
+------+-------+
|    1 | Achal |
+------+-------+
1 row in set (0.00 sec)

mysql> |
```

-Create a backup dirrectory

```
root@LAPTOP-HPUOTJF1:~# mkdir ~/backup
root@LAPTOP-HPUOTJF1:~# docker run --rm -v mysql-data:/data -v ~/backup:/backup ubuntu tar czvf /backup/mysql_backup.tar.gz -C /data .
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
a3679419df18: Pull complete
ed819469700f: Pull complete
e16351a257e4: Download complete
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Downloaded newer image for ubuntu:latest
```

-copying the data into host dirrectory

```
./performance_schema/replication_appl_167.sdi
./performance_schema/events_statement_127.sdi
./performance_schema/events_stages_hi_112.sdi
./undo_001
./auto.cnf
./ib_buffer_pool
./server-cert.pem
./client-cert.pem
root@LAPTOP-HPUOTJF1:~# ls ~/backup
mysql_backup.tar.gz
root@LAPTOP-HPUOTJF1:~# |
```

- Restore the data to a new container and verify that the restoration was successful.

```
root@LAPTOP-HPUOTJF1:~# docker stop mysql-db
mysql-db
root@LAPTOP-HPUOTJF1:~# docker rm mysql-db
mysql-db
root@LAPTOP-HPUOTJF1:~# docker volume rm mysql-data
mysql-data
root@LAPTOP-HPUOTJF1:~# docker volume create mysql-data-new
mysql-data-new
root@LAPTOP-HPUOTJF1:~# docker run --rm -v mysql-data-new:/restore -v ~/backup:/backup ubuntu bash -c "cd /restore && tar xzvf /backup/mysql_backup.tar.gz"
./
./undo_002
./binlog.index
./mysql/
./mysql/slow_log.CSM
./mysql/general_log.CSV
./mysql/general_log.CSM
./mysql/slow_log.CSV
./mysql/general_log_213.sdi
```

- Start MySQL using the restored volume

```
./client-cert.pem
root@LAPTOP-HPUOTJF1:~# docker run -d --name mysql-new -v mysql-data-new:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=root123 -p 3308:3306 mysql:8.0
7ef1b03d5e5538c78af272252821ac5e9af9d012c59b74aee5e45eea89685881
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE       COMMAND                  CREATED            STATUS            PORTS                                                   NAMES
7ef1b03d5e55   mysql:8.0   "docker-entrypoint.s…"   4 seconds ago      Up 3 seconds      33060/tcp, 0.0.0.0:3308->3306/tcp, [::]:3308->3306/tcp   mysql-new
012bbee786a4   nginx       "/docker-entrypoint.…"   About an hour ago  Up About an hour  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp                  nginx-bind
root@LAPTOP-HPUOTJF1:~# |
```

- Verify the restore

```
root@LAPTOP-HPUOTJF1:~# docker run -d --name mysql-new -v mysql-data-new:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=root123 -p 3308:3306 mysql:8.0
7ef1b03d5e5538c78af272252821ac5e9af9d012c59b74aee5e45eea89685881
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE       COMMAND                  CREATED            STATUS            PORTS                                                   NAMES
7ef1b03d5e55   mysql:8.0   "docker-entrypoint.s…"   4 seconds ago      Up 3 seconds      33060/tcp, 0.0.0.0:3308->3306/tcp, [::]:3308->3306/tcp   mysql-new
012bbee786a4   nginx       "/docker-entrypoint.…"   About an hour ago  Up About an hour  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp                  nginx-bind
root@LAPTOP-HPUOTJF1:~# docker exec -it mysql-new mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE company;
LECT * FROM employee;Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql>
mysql> SELECT * FROM employee;
+------+-------+
| id   | name  |
+------+-------+
|    1 | Achal |
+------+-------+
1 row in set (0.00 sec)

mysql> |
```

40. **Docker Compose networking**: Create a Docker Compose file that defines a multi-container application with a frontend, backend, and database. Set up custom networks and volumes for the services and ensure that they can communicate with each other and store data persistently

-Create a Docker Compose file that defines a multi-container application with a frontend, backend, and database.

```
root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# mkdir -p compose-demo/frontend compose-demo/backend
root@LAPTOP-HPUOTJF1:~# ls
1ff                   GitIgnore-Repo  backup             cloudethix-nginx     githubaction-project  nginx-container.tar  revert_demo   ubuntu-custom
Amend_repo            GitWorking      bind-mount-demo    compose-demo         httpd                 nginx-update         stash-demo    ubuntu-nginx
Cloud_Ethix_Copyfiles Merge_demo      ci-cd-project      docker-compose-demo  index.html            project              stash1-demo   volume-backup
DevOps_Project        amend-repo      clone-remote-repo  free-for-dev         merge_conflict_demo   revert-demo          student-repo
root@LAPTOP-HPUOTJF1:~# cd compose-demo/
root@LAPTOP-HPUOTJF1:~/compose-demo# ls
backend  frontend
root@LAPTOP-HPUOTJF1:~/compose-demo# cd frontend/
root@LAPTOP-HPUOTJF1:~/compose-demo/frontend# vim index.html
root@LAPTOP-HPUOTJF1:~/compose-demo/frontend# cd
root@LAPTOP-HPUOTJF1:~# cd compose-demo/backend/
root@LAPTOP-HPUOTJF1:~/compose-demo/backend# vim package.json
root@LAPTOP-HPUOTJF1:~/compose-demo/backend# vim server.js
root@LAPTOP-HPUOTJF1:~/compose-demo/backend# vim Dockerfile
root@LAPTOP-HPUOTJF1:~/compose-demo/backend# cd ..
root@LAPTOP-HPUOTJF1:~/compose-demo# vim docker-compose.yml
root@LAPTOP-HPUOTJF1:~/compose-demo# docker compose up
WARN[0000] /root/compose-demo/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
WARN[0000] Docker Compose is configured to build using Bake, but buildx isn't installed
[+] Building 62.9s (12/12) FINISHED
 => [backend internal] load build definition from Dockerfile
 => => transferring dockerfile: 151B
 => [backend internal] load metadata for docker.io/library/node:20
 => [backend auth] library/node:pull token for registry-1.docker.io
 => [backend internal] load .dockerignore
 => => transferring context: 2B
```

-docker-compose file

```yaml
version: "3.9"

services:

  frontend:
    image: nginx
    container_name: frontend
    ports:
      - "8080:80"
    volumes:
      - ./frontend:/usr/share/nginx/html
    networks:
      - cloud-network

  backend:
    build: ./backend
    container_name: backend
    ports:
      - "3000:3000"
    networks:
      - cloud-network

  database:
    image: mysql:8.0
    container_name: mysql-db
    environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: company
    volumes:
      - mysql-data:/var/lib/mysql
    ports:
      - "3307:3306"
    networks:
      - cloud-network

networks:
  cloud-network:

volumes:
  mysql-data:
~
~
~
~
~
~
~
~
```

```
backend exited with code 137
root@LAPTOP-HPUOTJF1:~/compose-demo# docker compose up -d
WARN[0000] /root/compose-demo/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 3/3
 ✔Container mysql-db  Started
 ✔Container frontend  Started
 ✔Container backend   Started
root@LAPTOP-HPUOTJF1:~/compose-demo# docker network ls
NETWORK ID     NAME                                DRIVER    SCOPE
14516bb6eaf3   bridge                              bridge    local
a964394174fa   cloud-network                       bridge    local
dad2a1b0db20   compose-demo_cloud-network          bridge    local
947f59718bf2   compose-project_default             bridge    local
3411a96dcd47   docker-compose-demo_cloud-network   bridge    local
20f475dfbeef   host                                host      local
f1eff302c182   none                                null      local
root@LAPTOP-HPUOTJF1:~/compose-demo# docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED              STATUS          PORTS                                                    NAMES
8e57474524d1   nginx                  "/docker-entrypoint.…"   About a minute ago   Up 52 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp                  frontend
6b1938d9fce3   mysql:8.0              "docker-entrypoint.s…"   About a minute ago   Up 52 seconds   33060/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp   mysql-db
a5f2d194c5f4   compose-demo-backend   "docker-entrypoint.s…"   About a minute ago   Up 52 seconds   0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp              backend
root@LAPTOP-HPUOTJF1:~/compose-demo#
```

- ensure that they can communicate with each other and store data persistently

```
        "com.docker.compose.network": "cloud-network",
        "com.docker.compose.project": "compose-demo",
        "com.docker.compose.version": "2.40.3"
    },
    "Containers": {
        "6b1938d9fce3e8c2c4d0fdc63dd960741edbe9861aff6e2d30f765c8865011ed": {
            "Name": "mysql-db",
            "EndpointID": "ca915b86c7ce4816a110cb0555aa69253a1b3c6322b2b85ac5bdd92fd2d6b632",
            "MacAddress": "3e:a9:8a:3f:45:87",
            "IPv4Address": "172.22.0.4/16",
            "IPv6Address": ""
        },
        "8e57474524d12f0351b316ce48802d42ff376c8f2d106850bf65db26e00023df": {
            "Name": "frontend",
            "EndpointID": "14d941d8b2761451f201a8cd40ffe86e6b01d3162bff0d6dade43b9e90815698",
            "MacAddress": "42:41:60:83:63:11",
            "IPv4Address": "172.22.0.2/16",
            "IPv6Address": ""
        },
        "a5f2d194c5f4774c230dff6234d46315b4f3e1e0b58ee38c3281696ceb56a58d": {
            "Name": "backend",
            "EndpointID": "84903c207e57f5b45d063090887666958f948a1b14867d499872669ec88c4e49",
            "MacAddress": "da:e8:db:20:4f:62",
            "IPv4Address": "172.22.0.3/16",
            "IPv6Address": ""
        }
    },
    "Status": {
```